
UNIFYING FORMAL EXPLANATIONS: A COMPLEXITY-THEORETIC PERSPECTIVE

Anonymous authors
Anonymous submission

ABSTRACT

Prior work has analyzed the computational complexity of constructing two canonical classes of explanations for ML model predictions: (i) *sufficient reasons*, defined as feature subsets whose fixed values determine a prediction, and (ii) *contrastive reasons*, defined as feature subsets whose modification changes a prediction. Existing results treat these objects under several distinct regimes, including non-probabilistic versus probabilistic semantics and local versus global explanation targets. This paper introduces a unified formal framework in which all of these explanation variants are expressed as minimization problems over a single probabilistic value function. We prove that the computational behavior of these minimization problems is governed by three structural properties of the value function: (i) *monotonicity*, (ii) *submodularity*, and (iii) *supermodularity*, which are fundamental notions in *combinatorial optimization*. The resulting characterization yields several technically non-obvious separations. In particular, although the *local* value functions fail to satisfy monotonicity and also fail to satisfy the relevant submodularity or supermodularity conditions, the corresponding *global* value functions do satisfy these properties. This structural separation yields new polynomial-time algorithms for computing multiple globally defined explanations with provable guarantees across model classes spanning the interpretability spectrum, including neural networks, decision trees, and tree ensembles. Conversely, we prove that even substantially simplified local variants of the same explanation tasks are NP-hard.

1 INTRODUCTION

Despite substantial progress in methods for explaining ML model decisions, a persistent theoretical obstacle is that many explanation notions equipped with formal guarantees are computationally difficult to construct (Barceló et al., 2020; Van den Broeck et al., 2022). This difficulty is typically amplified for complex or highly non-linear model classes (Barceló et al., 2020; Adolfi et al., 2025; 2024). Consequently, the *computational complexity* of explanation generation has become a central object of formal analysis: recent work seeks to classify which explanation guarantees admit efficient algorithms for which model classes, and which guarantees remain computationally inaccessible (Barceló et al., 2020; Wäldchen et al., 2021; Arenas et al., 2022; 2023; Marzouk & De La Higuera, 2024; Ordyniak et al., 2023; Laber, 2024; Bhattacharjee & Luxburg, 2024; Blanc et al., 2021; 2022).

From sufficient to contrastive reasons. Within the complexity-theoretic study of explanation generation, two explanation objects have received sustained attention: (i) *sufficient reasons* and (ii) *contrastive reasons* (Barceló et al., 2020; Arenas et al., 2022; Audemard et al., 2022a; Arenas et al., 2021; Barceló et al., 2025; Marques-Silva & Ignatiev, 2022; Ignatiev et al., 2020b; Darwiche & Hirth, 2020). A *sufficient reason* is a feature subset S such that, after the coordinates in S are fixed to specified values, the model output is invariant to all assignments of the complementary coordinates $\bar{S}$. A *contrastive reason* is a feature subset S whose modification can induce a change in the model prediction.

In contrast to additive attribution methods, which distribute scalar importance scores over features but can be difficult for humans to interpret (Kumar et al., 2020) or may not support actionable interventions (Bilodeau et al., 2024), sufficient and contrastive reasons specify *discrete, condition-like explanations*. They therefore encode answers to questions of the form “which feature conditions are sufficient for this prediction?” and “which feature changes can alter it?”. This discrete structure

explains their role in classical XAI methods (Ribeiro et al., 2018; Carter et al., 2019; Ignatiev et al., 2019; Dhurandhar et al., 2018), their empirical advantages for improving human prediction relative to additive models (Ribeiro et al., 2018; Yin & Neubig, 2022; Dhurandhar et al., 2018), and their utility in downstream tasks such as bias detection (Balkir et al., 2022; Carter et al., 2021; La Malfa et al., 2021; Muthukumar et al., 2018), model debugging (Jacovi et al., 2021), and anomaly detection (Davidson et al., 2025). A standard interpretability criterion is that smaller sufficient or contrastive feature sets are preferable; accordingly, *minimality* is treated as a central formal guarantee (Ribeiro et al., 2018; Lopardo et al., 2023; Carter et al., 2019; Barceló et al., 2020; Arenas et al., 2022; Blanc et al., 2021; Wäldchen et al., 2021).

Barceló et al. (2020) conducted one of the earliest studies on the complexity of deriving sufficient and contrastive reasons. Their work established that finding the minimal-sized sufficient reason for a decision tree is NP-hard, with the complexity further increasing to Σ_2^P -hard for neural networks. In the case of contrastive reasons, they demonstrated that computing the smallest possible explanation is solvable in polynomial time for decision trees but becomes NP-Hard for neural networks. Similar hardness results were later shown to hold for tree ensembles as well (Izza & Marques-Silva, 2021; Ordyniak et al., 2024; Audemard et al., 2022b).

From non-probabilistic to probabilistic explanations. A common limitation of the classical sufficient- and contrastive-reason definitions is that their universal quantification over domains makes them rigid: the required condition must hold absolutely, which can force explanations to include many features and thereby reduce informativeness (Ignatiev et al., 2019; 2020a; Arenas et al., 2022; Wäldchen et al., 2021). This limitation motivates probabilistic generalizations of sufficient and contrastive reasons (Wäldchen et al., 2021; Arenas et al., 2022; Blanc et al., 2021; Izza et al., 2023; Xue et al., 2023). Under these definitions, the relevant sufficiency or contrastivity condition is required to hold with probability above a prescribed threshold δ , rather than for every possible completion or intervention.

Wäldchen et al. (2021) initiated the complexity analysis of probabilistic sufficient reasons and proved NP^{PP} -hardness for CNF classifiers. The same hardness transfers to tree ensembles and neural networks (Barceló et al., 2020; Ordyniak et al., 2023). A key structural phenomenon in the probabilistic regime is that the induced probability function is not monotone (Arenas et al., 2022; Izza et al., 2023); this failure of monotonicity is sufficient to make even *subset minimal* explanations computationally difficult. In particular, (Arenas et al., 2022) prove that computing a subset minimal probabilistic sufficient reason is NP-hard even for decision trees, whereas the corresponding non-probabilistic explanations are computable in polynomial time (Huang et al., 2021; Barceló et al., 2020).

From local to global explanations. More recently, Bassan et al. (2024) extended the complexity-theoretic treatment from the *local* non-probabilistic regime, where an explanation is tied to one input prediction, to the *global* non-probabilistic regime, where sufficient or contrastive reasons are required to hold over an entire domain. This global universal requirement is at least as restrictive as its local analogue and inherits the rigidity of other non-probabilistic formalisms (Ignatiev et al., 2019; Barceló et al., 2020; Arenas et al., 2021; Darwiche & Hirth, 2020). For example, every feature excluded from a global sufficient reason is certified as strictly redundant (Bassan et al., 2024); consequently, the resulting explanation may contain nearly all input features and thus provide limited explanatory compression.

OUR CONTRIBUTIONS

1. We formulate prior explanation-computation problems — including sufficient, contrastive, probabilistic, non-probabilistic, local, and global variants — as instances of a single minimization problem over a unified value function. We then isolate three structural properties of this value function that determine the algorithmic status of the task: (a) *monotonicity*, (b) *supermodularity*, and (c) *submodularity*.
2. We prove that these properties depend sharply on whether the value function is local or global. Specifically, the *local* value functions for both sufficient and contrastive reasons are *non-monotonic*, whereas the corresponding *global* value functions are *monotonic non-decreasing*. We further establish global-only second-order structure: the global sufficient

value function is *supermodular*, while the global contrastive value function is *submodular*; neither property is present in the corresponding local setting.

3. We use this structural characterization to derive complexity results for explanation generation across three model families spanning the interpretability spectrum: (i) neural networks, (ii) decision trees, and (iii) tree ensembles. First, although subset-minimal local probabilistic sufficient/contrastive explanations are NP-hard to compute even for decision trees (Arenas et al., 2022), we prove that their global counterparts can be computed in polynomial time. We then extend this positive result to arbitrary black-box models, including neural networks and tree ensembles, under empirical distributions; in these cases, subset-minimal global sufficient/contrastive explanations are polynomial-time computable, whereas the corresponding local tasks remain NP-hard.
4. Finally, we exploit the submodular and supermodular structure of the global value functions to obtain *constant-factor* approximation guarantees for cardinally minimal global explanations, including for complex model classes such as neural networks and tree ensembles under fixed empirical distributions. The analogous local problems admit strong inapproximability: no *bounded* approximation exists even under very restricted assumptions.

The main text states the principal theorems and corollaries, while the appendix contains the complete formal proofs.

2 PRELIMINARIES

Setting. Let the input dimension be $n \in \mathbb{N}$ and write input vectors as $\mathbf{x} := (\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n)$. Coordinate $\mathbf{x}_i$ ranges over a domain $\mathcal{X}_i$, which may be discrete or continuous, and the full feature space is $\mathbb{F} := \mathcal{X}_1 \times \mathcal{X}_2 \times \dots \times \mathcal{X}_n$. We study classifiers $f : \mathbb{F} \rightarrow [c]$, where $c \in \mathbb{N}$ denotes the number of classes. We also assume an input distribution $\mathcal{D} : \mathbb{F} \rightarrow [0, 1]$. Because exact access to, or accurate approximation of, $\mathcal{D}$ may be computationally infeasible, we also consider a fixed empirical dataset $\mathbf{D} := \{\mathbf{z}^1, \mathbf{z}^2, \dots, \mathbf{z}^{|\mathbf{D}|}\}$ as a finite proxy for the underlying data-generating distribution, for example when $\mathbf{D}$ is sampled from the available training data.

We distinguish between local and global explanation targets. A local explanation is conditioned on a fixed input $\mathbf{x} \in \mathbb{F}$ and explains the prediction $f(\mathbf{x})$. A global explanation is not tied to a single input; instead, it specifies feature conditions that characterize the behavior of f over the relevant input domain or distribution.

Models. Several of our results are model-independent structural statements about value functions, but we also instantiate the framework to obtain model-specific complexity classifications. To cover both conventionally interpretable and black-box regimes, we analyze: (i) decision trees, (ii) neural networks (all architectures at least as expressive as feed-forward ReLU networks), and (iii) tree ensembles, including majority-voting ensembles (e.g., random forests) and weighted-voting ensembles (e.g., XGBoost). The appendix gives the corresponding formal model definitions.

Distributions. In probabilistic explanation problems, the distributional representation is part of the computational input and can change the resulting complexity. We therefore separate three distribution classes: (i) *general distributions*, which impose no structural assumptions on $\mathcal{D}$ and are used for universally quantified value-function properties; (ii) *empirical distributions*, which are induced by a finite dataset $\mathbf{D}$ and are standard in XAI analyses (Lundberg & Lee, 2017; Van den Broeck et al., 2022); and (iii) *independent distributions*, which require mutual independence among features under $\mathcal{D}$ and are also common in the XAI literature (Arenas et al., 2022; 2023; Lundberg & Lee, 2017; Ribeiro et al., 2018). Empirical distributions need not satisfy feature independence; they may encode arbitrary dependencies present in the finite dataset (Van den Broeck et al., 2022). The appendix provides the full formal definitions.

3 FORMS OF REASONS

This section defines the explanation objects used throughout the paper. We first specify the non-probabilistic local and global variants of sufficient and contrastive reasons, and then formulate

their probabilistic counterparts by replacing universal requirements with thresholded probability conditions.

3.1 Non-Probabilistic SUFFICIENT AND CONTRASTIVE REASONS

Sufficient reasons. In feature-selection explanations, a subset of coordinates is selected as the relevant support for a model decision. We study the standard *sufficiency* criterion for such subsets, which underlies several explainability methods (Ribeiro et al., 2018; Carter et al., 2019; Ignatiev et al., 2019; Dasgupta et al., 2022). The criterion may be instantiated locally, for a fixed prediction, or globally, over the full input domain. Following the standard definition, a *local sufficient reason* for $\langle f, \mathbf{x} \rangle$ is a feature subset $S \subseteq \{1, \dots, n\}$ such that fixing the coordinates in S to their values in $\mathbf{x}$ forces the prediction to remain $f(\mathbf{x})$ for every assignment of the remaining coordinates $\bar{S}$. Formally, S is a local sufficient reason for $\langle f, \mathbf{x} \rangle$ iff:

$$\forall \mathbf{z} \in \mathbb{F}, \quad f(\mathbf{x}_S; \mathbf{z}_{\bar{S}}) = f(\mathbf{x}). \quad (1)$$

Here, $(\mathbf{x}_S; \mathbf{z}_{\bar{S}})$ denotes the hybrid vector whose coordinates in S are copied from $\mathbf{x}$ and whose coordinates in $\bar{S}$ are copied from $\mathbf{z}$. This definition is closely related to *semi-factual* explanations (Kenny & Keane, 2021; Alfano et al., 2025), which search for an alternative input $\mathbf{z}'$ satisfying $f(\mathbf{z}') = f(\mathbf{x})$ while remaining close to the original input. If $\mathbf{z}'$ is restricted to the form $(\mathbf{x}_S; \mathbf{z}_{\bar{S}})$ and closeness is measured by Hamming distance, the two notions coincide.

A *global sufficient reason* (Bassan et al., 2024) is a feature subset $S \subseteq \{1, \dots, n\}$ that satisfies the local sufficiency condition simultaneously for every input $\mathbf{x} \in \mathbb{F}$:

$$\forall \mathbf{x}, \mathbf{z} \in \mathbb{F}, \quad f(\mathbf{x}_S; \mathbf{z}_{\bar{S}}) = f(\mathbf{x}). \quad (2)$$

Contrastive reasons. A complementary explanation objective is to identify feature subsets whose intervention can *modify* a prediction (Dhurandhar et al., 2018; Mothilal et al., 2020; Guidotti, 2024). Such explanations formalize the coordinates on which a change may be sufficient to flip the model output. A subset $S \subseteq \{1, \dots, n\}$ is a *local contrastive reason* for $\langle f, \mathbf{x} \rangle$ iff:

$$\exists \mathbf{z} \in \mathbb{F}, \quad f(\mathbf{z}_S; \mathbf{x}_{\bar{S}}) \neq f(\mathbf{x}). \quad (3)$$

Contrastive reasons correspond to *counterfactual* explanations (Guidotti, 2024; Mothilal et al., 2020) when counterfactuals are restricted to nearby assignments $\mathbf{x}'$ satisfying $f(\mathbf{x}') \neq f(\mathbf{x})$ and distance is measured by Hamming weight.

The global contrastive analogue requires the same intervention capacity uniformly over the domain: changing coordinates in S must be able to alter the prediction for every input under consideration. This object is related to formal bias-identification methods (Arenas et al., 2021; Bassan et al., 2024; Darwiche & Hirth, 2020) and to *group* counterfactual methods that construct counterfactuals for multiple data instances (Carrizosa et al., 2024; Warren et al., 2024). Formally, S is a *global contrastive reason* for f iff:

$$\forall \mathbf{x} \in \mathbb{F}, \quad \exists \mathbf{z} \in \mathbb{F}, \quad f(\mathbf{z}_S; \mathbf{x}_{\bar{S}}) \neq f(\mathbf{x}). \quad (4)$$

3.2 Probabilistic SUFFICIENT AND CONTRASTIVE REASONS

As noted above, non-probabilistic sufficient and contrastive reasons impose universal conditions that are often too stringent. Probabilistic definitions (Arenas et al., 2022; Ribeiro et al., 2018; Izza et al., 2023; Wäldchen et al., 2021; Bounia & Koriche, 2023; Subercaseaux et al., 2025; Wang et al., 2021a; Blanc et al., 2021) relax these conditions by requiring the relevant event to occur with probability at least $\delta \in [0, 1]$. The case $\delta = 1$ exactly recovers the corresponding non-probabilistic definitions.

Probabilistic sufficient reasons. A subset $S \subseteq \{1, \dots, n\}$ is a *local δ -sufficient reason* for $\langle f, \mathbf{x} \rangle$ when conditioning the coordinates in S to match $\mathbf{x}$ and sampling the remaining coordinates according to $\mathcal{D}$ preserves the classification with probability at least δ . Equivalently:

$$\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S) \geq \delta. \quad (5)$$

The condition $\mathbf{z}_S = \mathbf{x}_S$ specifies that the coordinates of $\mathbf{z}$ indexed by S are fixed to the corresponding coordinates of $\mathbf{x}$. For global explanations, we aggregate the local sufficiency probability over inputs sampled from $\mathcal{D}$. Thus, $S \subseteq \{1, \dots, n\}$ is a *global δ -sufficient reason* for $\langle f \rangle$ if this expected local sufficiency probability is at least δ :

$$\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \geq \delta. \quad (6)$$

Probabilistic contrastive reasons. The probabilistic local contrastive condition fixes the complementary coordinates $\bar{S}$ to their values in $\mathbf{x}$ and samples the coordinates in S . The subset S is a *local δ -contrastive reason* if the sampled intervention changes the original prediction $f(\mathbf{x})$ with probability at least δ . Formally:

$$\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) \neq f(\mathbf{x}) \mid \mathbf{z}_{\bar{S}} = \mathbf{x}_{\bar{S}}) \geq \delta. \quad (7)$$

For the global contrastive setting, we again average the corresponding local probability over inputs drawn from $\mathcal{D}$. A subset S is a *global δ -contrastive reason* when this expected contrastivity probability is at least δ :

$$\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) \neq f(\mathbf{x}) \mid \mathbf{z}_{\bar{S}} = \mathbf{x}_{\bar{S}})] \geq \delta. \quad (8)$$

4 FORMS OF MINIMALITY

For all explanation variants considered above, smaller feature sets are typically preferred because they provide stronger compression of the model behavior. We therefore analyze two standard minimality notions, both applicable to sufficient and contrastive explanations in local, global, probabilistic, and non-probabilistic regimes:

Definition 1. Assuming a subset $S \subseteq \{1, \dots, n\}$ is an explanation, then:

1. S is a *cardinally-minimal explanation* (Barceló et al., 2020; Bassan et al., 2024) iff S has the smallest explanation cardinality $|S|$ (i.e., there is no explanation S' such that $|S'| < |S|$).
2. $S \subseteq \{1, \dots, n\}$ is a *subset-minimal explanation* (Arenas et al., 2022; Ignatiev et al., 2019) iff S is an explanation, and any $S' \subsetneq S$ is not an explanation.

Cardinal minimality is strictly stronger than subset minimality: every cardinally minimal explanation is subset-minimal, but the converse need not hold (Appendix B.3 gives an example). We use the terms subset-minimal and cardinally minimal rather than local and global minima to avoid ambiguity with local versus global explanation targets. Both minimality notions apply uniformly to all explanation forms studied here. For example, a cardinally minimal local probabilistic δ sufficient reason minimizes $|S|$ among all qualifying subsets, whereas a subset-minimal one only requires that no proper subset $S' \subsetneq S$ also qualifies.

5 A UNIFIED COMBINATORIAL OPTIMIZATION TASK

All explanation-computation problems introduced above can be represented as thresholded subset-selection problems: find a minimal subset S such that $v(S) \geq \delta$, with the non-probabilistic regime obtained by setting $\delta := 1$. We use v_{suff}^ℓ for the local sufficiency probability in equation 5, i.e., $\Pr_{\mathbf{z} \sim \mathcal{D}_p}(f(\mathbf{x}) = f(\mathbf{z}) \mid \mathbf{x}_S = \mathbf{z}_S)$, and v_{suff}^g for the global sufficiency probability in equation 6. Analogously, v_{con}^ℓ and v_{con}^g denote the local and global contrastive probabilities from equations 7 and 73. With this notation, the cardinally minimal δ -explanation task is:

Cardinally Minimal δ -Explanation:

Input: Model f , a distribution $\mathcal{D}$, (possibly, an input $\mathbf{x}$), a *general* value function $v : 2^n \rightarrow [0, 1]$ (defined using f , $\mathcal{D}$, and possibly $\mathbf{x}$), and some $\delta \in [0, 1]$.

Output: A subset $S \subseteq [n]$ such that $v(S) \geq \delta$ and $|S|$ is minimal.

Replacing cardinal optimality by inclusion-wise minimality yields the following relaxed optimization problem:

Subset Minimal δ -Explanation:

Input: Model f , a distribution $\mathcal{D}$, (possibly, an input $\mathbf{x}$), a general value function $v : 2^n \rightarrow [0, 1]$ (defined using f , $\mathcal{D}$, and possibly $\mathbf{x}$), and some $\delta \in [0, 1]$.

Output: A subset $S \subseteq [n]$ such that $v(S) \geq \delta$ and for any $S' \subseteq S$ it holds that $v(S') < \delta$.

Properties of v that affect the complexity. The algorithmic behavior of the preceding tasks depends on structural properties of the value function. The first such property is *non-decreasing monotonicity*, which requires every single-coordinate augmentation to have non-negative marginal value $v(S \cup \{i\}) - v(S)$. Formally:

Definition 2. We say that a value function v maintains non-decreasing monotonicity if for any $S \subseteq \{1, \dots, n\}$ and any $i \in \{1, \dots, n\}$ it holds that: $v(S \cup \{i\}) \geq v(S)$.

The second-order properties used in our analysis are *supermodularity* and *submodularity*. These conditions specify how the marginal gain $v(S \cup \{i\}) - v(S)$ changes as the base set grows: supermodularity requires increasing marginal gains, whereas submodularity requires decreasing marginal gains. Formally:

Definition 3. Let there be some value function v , some $S \subseteq S' \subseteq \{1, \dots, n\}$, and $i \notin S'$. Then:

1. v maintains supermodularity iff it holds that: $v(S \cup \{i\}) - v(S) \leq v(S' \cup \{i\}) - v(S')$.
2. v maintains submodularity iff it holds that: $v(S \cup \{i\}) - v(S) \geq v(S' \cup \{i\}) - v(S')$.

6 UNRAVELING NEW PROPERTIES OF THE GLOBAL VALUE FUNCTIONS

In the local non-probabilistic setting, subset-minimal sufficient and contrastive reasons are computable because the associated value function is monotone; however, this monotonicity is tied to the restrictive threshold $\delta = 1$. Allowing arbitrary probabilistic thresholds δ would increase modeling flexibility, but prior work shows that monotonicity fails in the local probabilistic regime, which substantially increases the difficulty of explanation computation (Arenas et al., 2022; Kozachinskiy, 2023; Izza et al., 2023; Subercaseaux et al., 2025; Izza et al., 2024b).

In the *global* non-probabilistic setting, Bassan et al. (2024) establish an even stronger structural property: there is exactly *one* subset-minimal explanation. This uniqueness result is again a consequence of the highly restrictive $\delta = 1$ semantics, especially because the global condition must hold for *all* inputs. Indeed, Bassan et al. (2024) show that the unique minimal subset is precisely the set of all features that are not strictly redundant, which may coincide with the full feature set. We prove that this uniqueness property does *not* extend to the general global *probabilistic* setting; for arbitrary δ , the number of subset-minimal explanations can be *exponential*.

Proposition 1. While the non-probabilistic case ($\delta = 1$) admits a unique subset-minimal global sufficient reason (Bassan et al., 2024), in the general probabilistic setting (for arbitrary δ), there exist functions f that have $\Theta(\frac{2^n}{\sqrt{n}})$ subset-minimal global sufficient reasons.

Although uniqueness is lost once arbitrary δ values are allowed, the global probabilistic value functions retain a different structural property that is algorithmically decisive: *monotonicity*. We prove that both the global sufficient value function v_{suff}^g and the global contrastive value function v_{con}^g are monotone for every δ , in direct contrast to the local value functions v_{suff}^ℓ and v_{con}^ℓ :

Proposition 2. While the local probabilistic setting (for any δ) lacks monotonicity — i.e., the value functions v_{con}^ℓ and v_{suff}^ℓ are non-monotonic (Arenas et al., 2022; Izza et al., 2023; Subercaseaux et al., 2025; Izza et al., 2024b) — in the global probabilistic setting (also for any δ), both value functions v_{con}^g and v_{suff}^g are monotonic non-decreasing.

Beyond monotonicity, the global value functions also satisfy second-order combinatorial structure absent from their local counterparts. Under feature independence, the global sufficient value function v_{suff}^g is supermodular. The corresponding local function v_{suff}^ℓ is not supermodular even under the stronger assumption that the input distribution is uniform, and hence independent. Specifically:

Proposition 3. While the local probabilistic sufficient setting (for any δ) lacks supermodularity — even when $\mathcal{D}$ is uniform, i.e., the value function v_{suff}^ℓ is not supermodular — in the global probabilistic setting (also for any δ), when $\mathcal{D}$ exhibits feature independence, the value function v_{suff}^g is supermodular.

For the global probabilistic *contrastive* objective, the second-order structure has the dual form: the value function is submodular rather than supermodular. This again separates the global regime from the local one, where the corresponding value function is neither submodular nor supermodular.

Proposition 4. *While the local probabilistic contrastive setting (for any δ) lacks submodularity — even when $\mathcal{D}$ is uniform, i.e., the value function v_{con}^ℓ is not submodular — in the global probabilistic setting (also for any δ), when $\mathcal{D}$ exhibits feature independence, the value function v_{con}^g is submodular.*

7 COMPUTATIONAL COMPLEXITY RESULTS

7.1 SUBSET MINIMAL EXPLANATIONS

This subsection analyzes *subset minimal* explanations across the local/global and sufficient/contrastive variants for the model classes under consideration. The decisive structural assumption is monotonicity of the value function. The preceding section proves monotonicity for the global functions v_{con}^g and v_{suff}^g , and recalls its failure for the local functions v_{con}^ℓ and v_{suff}^ℓ . This distinction determines whether a deletion-based greedy procedure is guaranteed to terminate at a subset-minimal explanation. We therefore obtain a sharp contrast: several local subset-minimal explanation tasks are computationally hard, whereas their global counterparts are polynomial-time solvable. We begin with the following generalized greedy algorithm:

Algorithm 1 Subset Minimal Explanation Search

Input Value function v , and some $\delta \in [0, 1]$

```

1:  $S \leftarrow \{1, \dots, n\}$ 
2: while  $\min_{i \in S} v(S \setminus \{i\}) \geq \delta$  do
3:    $j \leftarrow \operatorname{argmax}_{i \in S} v(S \setminus \{i\})$ 
4:    $S \leftarrow S \setminus \{j\}$ 
5: end while
6: return  $S$ 

```

▷ S is a (subset minimal?) δ -explanation

Algorithm 1 is a top-down search procedure for a subset-minimal δ -explanation. It initializes S to the full feature set $\{1, \dots, n\}$ and repeatedly tests whether deleting one coordinate can preserve the threshold constraint. At each iteration it removes a coordinate j that maximizes the residual value $v(S \setminus \{j\})$, i.e., a deletion that decreases the value function as little as possible among admissible single-coordinate deletions. The algorithm terminates when no single deletion preserves $v(S) \geq \delta$.

The correctness of this procedure as a subset-minimality algorithm depends on monotonicity of v . Termination only certifies that every single deletion from the returned set violates the threshold. If v is monotone, then every proper subset is bounded above by one of these single-deletion subsets, so the returned set is inclusion-wise minimal. Applying the monotonicity results from the previous section yields:

Proposition 5. *Computing Algorithm 1 with the local value functions v_{con}^ℓ and v_{suff}^ℓ does not always converge to a subset minimal δ -sufficient/contrastive reason. However, computing it with the global value functions v_{con}^g or v_{suff}^g necessarily produces subset minimal δ -sufficient/contrastive reasons.*

We next instantiate this algorithmic principle to obtain model-specific complexity classifications for subset-minimal explanations.

Decision trees. We first consider a restricted model class that is usually regarded as interpretable: decision trees under independent input distributions. Even in this setting, Arenas et al. (2022) prove that computing a subset-minimal local δ -sufficient reason is NP-hard unless PTIME=NP, including under the uniform distribution. We prove that the corresponding global task has the opposite complexity status: a subset-minimal global δ -sufficient reason for a decision tree is computable in polynomial time whenever the relevant probability term is efficiently computable. Formally:

Theorem 1. *If f is a decision tree and the probability term v_{suff}^g can be computed in polynomial time given the distribution $\mathcal{D}$ (which holds for independent distributions, among others), then obtaining a subset-minimal global δ -sufficient reason can be obtained in polynomial time. However, unless PTIME=NP, no polynomial-time algorithm exists for computing a local δ -sufficient reason for decision trees even under independent distributions.*

Extension to other tractable models. The same global tractability mechanism also applies to *orthogonal DNFs* (Crama & Hammer, 2011), i.e., Disjunctive Normal Form formulas whose terms are pairwise mutually exclusive. This extension is useful because orthogonal DNFs *generalize* decision trees while retaining a similarly tractable decomposition structure. Appendix N gives the full argument.

Neural networks, tree ensembles, and other complex models. We next extend the positive global result beyond decision trees. Under an empirical distribution, and assuming that the model f supports polynomial-time inference, the value of a candidate subset can be evaluated efficiently. Consequently, subset-minimal global δ -sufficient and δ -contrastive reasons are computable in polynomial time.

Proposition 6. *For any model f , and empirical distribution $\mathcal{D}$ — computing a subset-minimal global δ -sufficient or δ -contrastive reason for f can be done in polynomial time.*

Because this positive result holds for any polynomial-time inference model under an empirical distribution, it yields a direct separation from the local regime. For neural networks and tree ensembles, subset-minimal *local* explanations remain intractable even under empirical distributions. This gives the following strict complexity separation:

Theorem 2. *Assuming f is a neural network or a tree ensemble, and $\mathcal{D}$ is an empirical distribution — there exist polynomial-time algorithms for obtaining subset minimal global δ -sufficient and contrastive reasons. However, unless $\text{PTIME}=\text{NP}$, there is no polynomial time algorithm for computing a subset minimal local δ -sufficient reason or a subset minimal local δ -contrastive reason.*

7.2 APPROXIMATE CARDINALLY MINIMAL EXPLANATIONS

This subsection turns from inclusion-wise minimality to the stronger objective of computing *cardinally minimal* δ -sufficient and δ -contrastive reasons. In the global regime, monotonicity together with submodularity or supermodularity permits approximation guarantees for this optimization problem. In the local regime, the analogous tasks remain intractable and, in the cases below, inapproximable. Thus the separation between local and global explanation computation persists under the stricter cardinality objective.

A unified greedy approximation algorithm. For a non-decreasing submodular value function, the cardinal-minimal explanation problem is an instance of *submodular set cover* (Wolsey, 1982). For a supermodular value function, it becomes a related non-submodular set-cover variant (Shi et al., 2021). Both optimization problems admit known approximation analyses (Wolsey, 1982; Iyer & Bilmes, 2013; Chen & Crawford, 2023). We therefore use the standard greedy construction in Algorithm 2 as a common approximation template and then instantiate its guarantee separately for the contrastive and sufficient objectives.

Algorithm 2 Cardinally Minimal Explanation Approximation Search

Input Value function v , and some $\delta \in [0, 1]$

```

1:  $S \leftarrow \emptyset$ 
2: while  $\max_{i \in S} v(S \cup \{i\}) < \delta$  do
3:    $j \leftarrow \operatorname{argmax}_{i \in S} v(S \cup \{i\})$ 
4:    $S \leftarrow S \cup \{j\}$ 
5: end while
6: return  $S$  ▷  $|S|$  is a (provable approximation?) of a cardinally minimal  $\delta$ -explanation

```

Algorithm 2 is the bottom-up analogue of Algorithm 1. It initializes S to the empty set and repeatedly adds the coordinate that maximizes the resulting value $v(S \cup i)$, terminating once the threshold δ is reached.

Cardinally minimal contrastive reasons. For global contrastive reasons, the value function is both monotone and submodular. The cardinal-minimal problem therefore reduces to classical *submodular set cover*, so Algorithm 2 inherits the approximation guarantee of Wolsey (1982). For integer-valued objectives this gives a Harmonic-factor bound, and in general it gives an $\mathcal{O}\left(\ln\left(\frac{v([n])}{\min_{i \in [n]} v(i)}\right)\right)$ -approximation. Under an *empirical* distribution $\mathcal{D}$ of fixed sample size, the approximation factor is

constant and only logarithmic in the sample size. In contrast, the *local* problem admits no *bounded* approximation even for a *single* sample point.

Theorem 3. *Given a neural network or tree ensemble f and an empirical distribution $\mathcal{D}$ over a fixed dataset D , Algorithm 2 yields a constant $\mathcal{O}\left(\ln\left(\frac{v_{\text{con}}^g([n])}{\min_{i \in [n]} v_{\text{con}}^g(\{i\})}\right)\right)$ -approximation, bounded by $\mathcal{O}(\ln(|D|))$, for computing a global cardinally minimal δ -contrastive reason for f , assuming feature independence. In contrast, unless $\text{PTIME}=\text{NP}$, no bounded approximation exists for computing a local cardinally minimal δ -contrastive reason for any $\langle f, \mathbf{x} \rangle$, even when $|D| = 1$.*

We also prove a matching *lower bound* for the guarantee in Theorem 3 for empirical distributions without an independence assumption; see Appendix M.

Cardinally minimal sufficient reasons. The sufficient-reason objective corresponds to a supermodular set-cover variant rather than the submodular set cover problem used for contrastive reasons. This variant is harder to approximate, but it admits guarantees when the value function has bounded curvature (Shi et al., 2021). For a function $v : 2^n \rightarrow \mathbb{R}$, the *total curvature* is $k^f := 1 - \min_{i \in [n]} \frac{v([n]) - v([n] \setminus i)}{v(i) - v(\emptyset)}$. Using the approximation result of Shi et al. (2021), we show that Algorithm 2 achieves an $\mathcal{O}\left(\frac{1}{1-k^f} + \ln\left(\frac{v([n])}{\min_{i \in [n]} v(i)}\right)\right)$ -approximation.

For a fixed empirical distribution, this approximation factor is constant. The sufficient-reason bound is weaker than the contrastive $\mathcal{O}(\ln(|D|))$ factor because it includes the additional $\frac{1}{1-k^f}$ term, but it remains a constant-factor guarantee. The local variant again has no *bounded* approximation, even when $|D| = 1$.

Theorem 4. *Given a neural network or tree ensemble f and an empirical distribution $\mathcal{D}$ over a fixed dataset D , Algorithm 2 yields a constant $\mathcal{O}\left(\frac{1}{1-k^f} + \ln\left(\frac{v_{\text{diff}}^g([n])}{\min_{i \in [n]} v_{\text{diff}}^g(\{i\})}\right)\right)$ -approximation for computing a global cardinally minimal δ -sufficient reason for f , assuming feature independence. In contrast, unless $\text{PTIME}=\text{NP}$, there is no bounded approximation for computing a local cardinally minimal δ -sufficient reason for any $\langle f, \mathbf{x} \rangle$, even when $|D| = 1$.*

Together with Subsection 7.1, these results show that the global/local separation is robust to the choice of minimality notion: subset-minimal global explanations are exactly computable, and cardinally minimal global explanations are approximable, whereas the corresponding local problems remain hard or inapproximable.

8 LIMITATIONS AND FUTURE WORK

Many of our central results, especially the structural statements about value functions, are stated at a general level. We then instantiate these statements to obtain concrete complexity classifications for particular model classes, distributional assumptions, and explanation definitions, including neural networks and empirical distributions. Other combinations of models, distributions, and explanation semantics remain outside the present analysis. Nevertheless, the results identify value-function structure as a mechanism underlying both tractability and intractability, and thereby provide a formal basis for analyzing additional explainability settings.

Several theoretical directions follow directly from this framework:

1. First, the positive complexity bounds for global explanations may be sharpenable for simpler model classes, including decision trees, linear models, and other representations with inherently tractable structure. Although local explanations for such classes can remain intractable (Arenas et al., 2022; Subercaseaux et al., 2025), the global objectives studied here have additional structure that may support stronger exact or approximation guarantees. Tightening the empirical-distribution approximation bounds is a particularly direct open problem.
2. Second, when *exact* expectation computation is infeasible, one can ask whether approximation procedures such as Fully Polynomial Randomized Approximation Schemes (FPRAS) for DNF-like model classes (Meel et al., 2019) (and therefore for tree ensembles) or Monte Carlo approximations of the type used in (Subercaseaux et al., 2025) can be integrated

into the framework. The main technical issue is to characterize which approximation-error tolerances preserve subset-minimality or cardinal-approximation guarantees.

3. Third, alternative feature-importance objectives, including loss-based criteria such as KL divergence (Conmy et al., 2023), may exhibit analogous monotonicity, submodularity, or supermodularity transitions between local and global formulations. Identifying such structure could yield efficient feature-selection algorithms beyond the sufficient/contrastive objectives analyzed here.

9 CONCLUSION

We developed a unified combinatorial framework for sufficient and contrastive explanations across local, global, probabilistic, and non-probabilistic regimes. The central technical distinction is structural: the *local* value functions lack the monotonicity, submodularity, and supermodularity properties needed for standard greedy guarantees, whereas the *global* value functions satisfy (i) *monotonicity*, (ii) *submodularity* for contrastive reasons, and (iii) *supermodularity* for sufficient reasons. These properties yield polynomial-time algorithms for subset-minimal global explanations and approximation guarantees for cardinally minimal global explanations, including for complex model classes such as neural networks and tree ensembles under the stated distributional assumptions. Conversely, the corresponding local tasks remain NP-hard or inapproximable even in simplified settings. The results therefore identify a formal mechanism by which global explanation objectives can be more tractable than their local analogues.

REFERENCES

- Federico Adolfi, Martina G Vilas, and Todd Wareham. Complexity-Theoretic Limits on the Promises of Artificial Neural Network Reverse-Engineering. In *Proc. 46th Annual Meeting of the Cognitive Science Society (CogSci)*, 2024.
- Federico Adolfi, Martina Vilas, and Todd Wareham. The Computational Complexity of Circuit Discovery for Inner Interpretability. In *Proc. 13th Int. Conf. on Learning Representations (ICLR)*, 2025.
- Gianvincenzo Alfano, Sergio Greco, Domenico Mandaglio, Francesco Parisi, Reza Shahbazian, and Irina Trubitsyna. Even-If Explanations: Formal Foundations, Priorities and Complexity. In *Proc. 39th AAAI Conf. on Artificial Intelligence*, pp. 15347–15355, 2025.
- Ibrahim M Almanjahie, Zouaoui Chikr Elmezouar, Ali Laksaci, and Mustapha Rachdi. KNN Local Linear Estimation of the Conditional Cumulative Distribution Function: Dependent Functional Data Case. *Comptes Rendus. Mathématique*, 356(10):1036–1039, 2018.
- David Alvarez Melis and Tommi Jaakkola. Towards Robust Interpretability with Self-Explaining Neural Networks. In *Proc. 32nd Conf. on Neural Information Processing Systems (NeurIPS)*, 2018.
- Leila Amgoud and Martin Cooper. Axiomatic Foundations of Counterfactual Explanations. 2026. Technical Report. <https://arxiv.org/abs/2602.04028>.
- Guy Amir, Shahaf Bassan, and Guy Katz. Hard to Explain: On the Computational Hardness of In-Distribution Model Interpretation. In *Proc. 27th European Conf. on Artificial Intelligence (ECAI)*, 2024.
- Alaa Anani, Tobias Lorenz, Mario Fritz, and Bernt Schiele. Pixel-level Certified Explanations via Randomized Smoothing. In *Proc. 42nd Int. Conf. on Machine Learning (ICML)*, 2025.
- Marcelo Arenas, D. Baez, P. Barceló, J. Pérez, and B. Subercaseaux. Foundations of Symbolic Languages for Model Interpretability. In *Proc. 34th Int. Conf. on Advances in Neural Information Processing Systems (NeurIPS)*, pp. 11690–11701, 2021.
- Marcelo Arenas, P. Barceló, M. Romero, and B. Subercaseaux. On Computing Probabilistic Explanations for Decision Trees. In *Proc. 35th Int. Conf. on Advances in Neural Information Processing Systems (NeurIPS)*, pp. 28695–28707, 2022.

540 Marcelo Arenas, Pablo Barceló, Leopoldo Bertossi, and Mikaël Monet. On the Complexity of SHAP-
541 Score-Based Explanations: Tractability via Knowledge Compilation and Non-Approximability
542 Results. *Journal of Machine Learning Research (JMLR)*, 24(63):1–58, 2023.

543 Gilles Audemard, S. Bellart, L. Bounia, F. Koriche, J. Lagniez, and P. Marquis. On the Computa-
544 tional Intelligibility of Boolean Classifiers. In *Proc. 18th Int. Conf. on Principles of Knowledge
545 Representation and Reasoning (KR)*, pp. 74–86, 2021.

546 Gilles Audemard, Steve Bellart, Louenas Bounia, Frédéric Koriche, Jean-Marie Lagniez, and Pierre
547 Marquis. On Preferred Abductive Explanations for Decision Trees and Random Forests. In *Proc.
548 31st Int. Joint Conf. on Artificial Intelligence (IJCAI)*, pp. 643–650, 2022a.

549 Gilles Audemard, Steve Bellart, Louenas Bounia, Frédéric Koriche, Jean-Marie Lagniez, and Pierre
550 Marquis. Trading Complexity for Sparsity in Random Forest Explanations. In *Proc. 36th AAAI
551 Conf. on Artificial Intelligence*, number 5, pp. 5461–5469, 2022b.

552 Gilles Audemard, Jean-Marie Lagniez, Pierre Marquis, and Nicolas Szczepanski. Computing
553 Abductive Explanations for Boosted Trees. In *Proc. 26th Int. Conf. on Artificial Intelligence and
554 Statistics (AISTATS)*, pp. 4699–4711, 2023.

555 Steve Azzolin, Sagar Malhorta, Andrea Passerini, and Stefano Teso. Beyond Topological Self-
556 Explainable GNNs: A Formal Explainability Perspective. In *Proc. 42nd Int. Conf. on Machine
557 Learning (ICML)*, 2025.

558 Esma Balkir, Isar Nejadgholi, Kathleen C Fraser, and Svetlana Kiritchenko. Necessity and Sufficiency
559 for Explaining Text Classifiers: A Case Study in Hate Speech Detection. In *Proc. Conf. of the
560 North American Chapter of the Association for Computational Linguistics: Human Language
561 Technologies (NAACL-HLT)*, pp. 2672–2686, 2022.

562 Pablo Barceló, M. Monet, J. Pérez, and B. Subercaseaux. Model Interpretability through the Lens of
563 Computational Complexity. In *Proc. 33rd Int. Conf. on Advances in Neural Information Processing
564 Systems (NeurIPS)*, pp. 15487–15498, 2020.

565 Pablo Barceló, Alexander Kozachinskiy, Miguel Romero, Bernardo Subercaseaux, and José Verschae.
566 Explaining k-Nearest Neighbors: Abductive and Counterfactual Explanations. In *Proc. 3rd ACM
567 on Management of Data*, pp. 1–26, 2025.

568 Shahaf Bassan and G. Katz. Towards Formal XAI: Formally Approximate Minimal Explanations
569 of Neural Networks. In *Proc. 29th Int. Conf. on Tools and Algorithms for the Construction and
570 Analysis of Systems (TACAS)*, pp. 187–207, 2023.

571 Shahaf Bassan, Guy Amir, Davide Corsi, Idan Refaeli, and Guy Katz. Formally Explaining Neural
572 Networks Within Reactive Systems. In *Proc. 23rd Int. Conf. on Formal Methods in Computer-Aided
573 Design (FMCAD)*, pp. 1–13, 2023.

574 Shahaf Bassan, Guy Amir, and Guy Katz. Local vs. Global Interpretability: A Computational
575 Complexity Perspective. In *Proc. 41st Int. Conf. on Machine Learning (ICML)*, 2024.

576 Shahaf Bassan, Guy Amir, Meirav Zehavi, and Guy Katz. What makes an Ensemble (Un) Inter-
577 pretable? In *Proc. 42nd Int. Conf. on Machine Learning (ICML)*, 2025a.

578 Shahaf Bassan, Yizhak Yisrael Elboher, Tobias Ladner, Matthias Althoff, and Guy Katz. Explaining,
579 Fast and Slow: Abstraction and Refinement of Provable Explanations. In *Proc. 42nd Int. Conf. on
580 Machine Learning (ICML)*, 2025b.

581 Shahaf Bassan, Ron Eliav, and Shlomit Gur. Explain Yourself, Briefly! Self-Explaining Neural
582 Networks with Concise Sufficient Reasons. In *Proc. 13th Int. Conf. on Learning Representations
583 (ICLR)*, 2025c.

584 Shahaf Bassan, Shlomit Gur, Sergey Zeltyn, Konstantinos Mavrogiorgos, Ron Eliav, and Dimosthenis
585 Kyriazis. Self-Explaining Neural Networks for Business Process Monitoring. 2025d. Technical
586 Report. <https://arxiv.org/abs/2503.18067>.

594 Shahaf Bassan, Michal Moshkovitz, and Guy Katz. Additive Models Explained: A Computational
595 Complexity Approach. In *Proc. 39th Conf. on Neural Information Processing Systems (NeurIPS)*,
596 2025e.

597 Shahaf Bassan, Yizhak Yisrael Elboher, Tobias Ladner, Volkan Şahin, Jan Kretinsky, Matthias Althoff,
598 and Guy Katz. Provably Explaining Neural Additive Models. In *Proc. 14th Int. Conf. on Learning*
599 *Representations (ICLR)*, 2026.

600 Robi Bhattacharjee and Ulrike Luxburg. Auditing Local Explanations is Hard. In *Proc. 38th Conf.*
601 *on Neural Information Processing Systems (NeurIPS)*, pp. 18593–18632, 2024.

602 Blair Bilodeau, Natasha Jaques, Pang Wei Koh, and Been Kim. Impossibility Theorems for Feature
603 Attribution. *Proc. of the National Academy of Sciences (PNAS)*, 121(2):e2304406120, 2024.

604 Guy Blanc, Jane Lange, and Li-Yang Tan. Provably Efficient, Succinct, and Precise Explanations.
605 In *Proc. 35th Int. Conf. on Advances in Neural Information Processing Systems (NeurIPS)*, pp.
606 6129–6141, 2021.

607 Guy Blanc, Caleb Koch, Jane Lange, and Li-Yang Tan. A Query-Optimal Algorithm for Finding
608 Counterfactuals. In *Proc. 39th Int. Conf. on Machine Learning (ICML)*, pp. 2075–2090, 2022.

609 Ryma Boumazouza, Fahima Cheikh-Alili, Bertrand Mazure, and Karim Tabia. ASTERYX: A model-
610 Agnostic SaT-basEd appRoach for SYmbolic and Score-Based EXplanations. In *Proc. 30th ACM*
611 *Int. Conf. on Information & Knowledge Management (CIKM)*, pp. 120–129, 2021.

612 Ryma Boumazouza, Raya Elsaleh, Melanie Ducoffe, Shahaf Bassan, and Guy Katz. FAME: For-
613 mal Abstract Minimal Explanation for Neural Networks. In *Proc. 14th Int. Conf. on Learning*
614 *Representations (ICLR)*, 2026.

615 Louenas Bounia. Enhancing the Intelligibility of Boolean Decision Trees with Concise and Reliable
616 Probabilistic Explanations. In *Proc. 20th Int. Conf. on Information Processing and Management of*
617 *Uncertainty in Knowledge-Based Systems (IPMU)*, pp. 205–218, 2024.

618 Louenas Bounia. Using Submodular Optimization to Approximate Minimum-Size Abductive Path
619 Explanations for Tree-Based Models. In *Proc. 41st Conf. on Uncertainty in Artificial Intelligence*
620 *(UAI)*, 2025.

621 Louenas Bounia and Frederic Koriche. Approximating Probabilistic Explanations via Supermodular
622 Minimization. In *Proc. 39th Conf. on Uncertainty in Artificial Intelligence (UAI)*, pp. 216–225,
623 2023.

624 Marco Calautti, Enrico Malizia, and Cristian Molinaro. On the Complexity of Global Necessary
625 Reasons to Explain Classification. 2025. Technical Report. [https://arXiv.org/abs/](https://arXiv.org/abs/2501.06766)
626 [2501.06766](https://arXiv.org/abs/2501.06766).

627 Emilio Carrizosa, Jasone Ramírez-Ayerbe, and Dolores Romero Morales. Mathematical Optimization
628 Modelling for Group Counterfactual Explanations. *European Journal of Operational Research*
629 *(EJOR)*, 319(2):399–412, 2024.

630 Brandon Carter, Jonas Mueller, Siddhartha Jain, and David Gifford. What Made You Do This?
631 Understanding Black-Box Decisions with Sufficient Input Subsets. In *Proc. 22nd Int. Conf. on*
632 *Artificial Intelligence and Statistics (AISTATS)*, pp. 567–576, 2019.

633 Brandon Carter, Siddhartha Jain, Jonas W Mueller, and David Gifford. Overinterpretation Reveals
634 Image Classification Model Pathologies. In *Proc. 35th Conf. on Neural Information Processing*
635 *Systems (NeurIPS)*, pp. 15395–15407, 2021.

636 Wenjing Chen and Victoria Crawford. Bicriteria Approximation Algorithms for the Submodular
637 Cover Problem. In *Proc. 37th Conf. on Neural Information Processing Systems (NeurIPS)*, pp.
638 72705–72716, 2023.

639 Arthur Conmy, Augustine Mavor-Parker, Aengus Lynch, Stefan Heimersheim, and Adrià Garriga-
640 Alonso. Towards Automated Circuit Discovery for Mechanistic Interpretability. *Proc. 37th Conf.*
641 *on Neural Information Processing Systems (NeurIPS)*, pp. 16318–16352, 2023.

648 Yves Crama and Peter L Hammer. *Boolean Functions: Theory, Algorithms, and Applications*.
649 Cambridge University Press, 2011.

650

651 Adnan Darwiche. Logic for Explainable AI. In *Proc. 38th ACM/IEEE Symposium on Logic in*
652 *Computer Science (LICS)*, pp. 1–11, 2023.

653 Adnan Darwiche and Auguste Hirth. On the Reasons Behind Decisions. In *Proc. 24th European*
654 *Conf. on Artificial Intelligence (ECAI)*, pp. 712–720, 2020.

655

656 Adnan Darwiche and Chunxi Ji. On the Computation of Necessary and Sufficient Explanations. In
657 *Proc. 36th AAAI Conf. on Artificial Intelligence*, pp. 5582–5591, 2022.

658 Sanjoy Dasgupta, Nave Frost, and Michal Moshkovitz. Framework for Evaluating Faithfulness of
659 Local Explanations. In *Proc. 39th Int. Conf. on Machine Learning (ICML)*, pp. 4794–4815, 2022.

660

661 Ian Davidson, Nicolás Kennedy, and SS Ravi. CXAD: Contrastive Explanations for Anomaly
662 Detection: Algorithms, Complexity Results and Experiments. *Transactions on Machine Learning*
663 *Research (TMLR)*, 2025.

664 Alessandro De Palma, Greta Dolcetti, and Caterina Urban. Faster Verified Explanations for Neural
665 Networks. 2025. Technical Report. <https://arXiv.org/abs/2512.00164>.

666

667 Amit Dhurandhar, Pin-Yu Chen, Ronny Luss, Chun-Chen Tu, Paishun Ting, Karthikeyan Shanmugam,
668 and Payel Das. Explanations Based on the Missing: Towards Contrastive Explanations with
669 Pertinent Negatives. *Proc. 32nd Conf. on Neural Information Processing Systems (NeurIPS)*, 2018.

670 Hidde Fokkema, Rianne De Heide, and Tim Van Erven. Attribution-Based Explanations that Provide
671 Recourse Cannot be Robust. *Journal of Machine Learning Research (JMLR)*, 24(360):1–37, 2023.

672

673 Hidde Fokkema, Damien Garreau, and Tim van Erven. The Risks of Recourse in Binary Classification.
674 In *Proc. 27th Int. Conf. on Artificial Intelligence and Statistics (AISTATS)*, pp. 550–558, 2024.

675 Tobias Geibinger, Reijo Jaakkola, Antti Kuusisto, Xinghan Liu, and Miikka Vilander. Why This and
676 Not That? A Logic-based Framework for Contrastive Explanations. In *Proc. 19th European Conf.*
677 *on Logics in Artificial Intelligence (JELIA)*, pp. 45–60. Springer, 2025.

678

679 Joshua Größer and Ostap Okhrin. Copulae: An Overview and Recent Developments. *Wiley Interdis-*
680 *ciplinary Reviews: Computational Statistics*, 14(3):e1557, 2022.

681 Riccardo Guidotti. Counterfactual Explanations and How to Find Them: Literature Review and
682 Benchmarking. *Data Mining and Knowledge Discovery (DMKD)*, 38(5):2770–2824, 2024.

683

684 Hyukjun Gweon, Matthias Schonlau, and Stefan H Steiner. The K Conditional Nearest Neighbor
685 Algorithm for Classification and Class Probability Estimation. *PeerJ Computer Science*, 5:e194,
686 2019.

687 Itamar Hadad, Shahaf Bassan, and Guy Katz. Formal Mechanistic Interpretability: Automated Circuit
688 Discovery with Provable Guarantees. In *Proc. 14th Int. Conf. on Learning Representations (ICLR)*,
689 2026.

690

691 Ouns El Harzli, Bernardo Cuenca Grau, and Ian Horrocks. Cardinality-Minimal Explanations for
692 Monotonic Neural Networks. In *Proc. 32nd Int. Joint Conf. on Artificial Intelligence (IJCAI)*, pp.
693 3677–3685, 2023.

694 W Keith Hastings. Monte Carlo Sampling Methods using Markov Chains and their Applications.
695 *Biometrika*, 57:97–109, 1970.

696

697 Rongjie Huang, Max W. Y. Lam, Jun Wang, Dan Su, Dong Yu, Yi Ren, and Zhou Zhao. FastDiff: A
698 Fast Conditional Diffusion Model for High-Quality Speech Synthesis. In *Proc. 31st Int. Joint Conf.*
699 *on Artificial Intelligence (IJCAI)*, pp. 4157–4163, 2022.

700 Xuanxiang Huang, Yacine Izza, Alexey Ignatiev, and Joao Marques-Silva. On Efficiently Explaining
701 Graph-Based Classifiers. In *Proc. 18th Int. Conf. on Principles of Knowledge Representation and*
Reasoning (KR), 2021.

-
- Alexey Ignatiev, Nina Narodytska, and Joao Marques-Silva. Abduction-Based Explanations for Machine Learning Models. In *Proc. 33rd AAAI Conference on Artificial Intelligence*, pp. 1511–1519, 2019.
- Alexey Ignatiev, Martin Cooper, Mohamed Siala, Emmanuel Hebrard, and Joao Marques-Silva. Towards Formal Fairness in Machine Learning. In *Proc. 26th Int. Conf. on Principles and Practice of Constraint Programming (CP)*, pp. 846–867, 2020a.
- Alexey Ignatiev, Nina Narodytska, Nicholas Asher, and Joao Marques-Silva. From Contrastive to Abductive Explanations and Back Again. In *Proc. Int. Conf. of the Italian Association for Artificial Intelligence (AIxAI)*, pp. 335–355, 2020b.
- Alexey Ignatiev, Yacine Izza, Peter J Stuckey, and Joao Marques-Silva. Using MaxSAT for Efficient Explanations of Tree Ensembles. In *Proc. 36th AAAI Conference on Artificial Intelligence*, pp. 3776–3785, 2022.
- Rishabh K Iyer and Jeff A Bilmes. Submodular Optimization with Submodular Cover and Submodular Knapsack Constraints. *Proc. 27th Conf. on Neural Information Processing Systems (NeurIPS)*, 2013.
- Yacine Izza and Joao Marques-Silva. On Explaining Random Forests with SAT. In *Proc. 30th Int. Joint Conf. on Artificial Intelligence (IJCAI)*, 2021.
- Yacine Izza, Xuanxiang Huang, Alexey Ignatiev, Nina Narodytska, Martin Cooper, and Joao Marques-Silva. On Computing Probabilistic Abductive Explanations. *Int. Journal of Approximate Reasoning (IJAR)*, 159:108939, 2023.
- Yacine Izza, Xuanxiang Huang, Antonio Morgado, Jordi Planes, Alexey Ignatiev, and Joao Marques-Silva. Distance-Restricted Explanations: Theoretical Underpinnings & Efficient implementation. In *Proc. 21st Int. Conf. on Principles of Knowledge Representation and Reasoning (KR)*, pp. 475–486, 2024a.
- Yacine Izza, Kuldeep S Meel, and Joao Marques-Silva. Locally-Minimal Probabilistic Explanations. In *Proc. 27th European Conf. on Artificial Intelligence (ECAI)*, 2024b.
- Reijo Jaakkola, Tomi Janhunen, Antti Kuusisto, Masood Feyzbakhsh Rankooh, and Miikka Vilander. Explainability via Short Formulas: the Case of Propositional Logic with Implementation. *Journal of Artificial Intelligence Research (JAIR)*, 83, 2025.
- Alon Jacovi, Swabha Swayamdipta, Shauli Ravfogel, Yanai Elazar, Yejin Choi, and Yoav Goldberg. Contrastive Explanations for Model Interpretability. In *Proc. Conf. Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pp. 1597–1611, 2021.
- Helen Jin, Anton Xue, Weiqiu You, Surbhi Goel, and Eric Wong. Probabilistic Stability Guarantees for Feature Attributions. In *Proc. 39th Conf. on Neural Information Processing Systems (NeurIPS)*, 2025.
- Amir-Hossein Karimi, Bernhard Schölkopf, and Isabel Valera. Algorithmic recourse: from counterfactual explanations to interventions. In *Proc. 4th ACM Conf. on Fairness, Accountability, and Transparency (FAccT)*, pp. 353–362, 2021.
- Eoin M Kenny and Mark T Keane. On Generating Plausible Counterfactual and Semi-Factual Explanations for Deep Learning. In *Proc. 35th AAAI Conf. on Artificial Intelligence*, number 13, pp. 11575–11585, 2021.
- Alexander Kozachinskiy. Inapproximability of Sufficient Reasons for Decision Trees. 2023. Technical Report. <https://arxiv.org/abs/2304.02781>.
- I Elizabeth Kumar, Suresh Venkatasubramanian, Carlos Scheidegger, and Sorelle Friedler. Problems with Shapley-Value-Based Explanations as Feature Importance Measures. In *Proc. 37th Int. Conf. on Machine Learning (ICML)*, pp. 5491–5500, 2020.

756 Emanuele La Malfa, Agnieszka Zbrzezny, Rhiannon Michelmore, Nicola Paoletti, and Marta
757 Kwiakowska. On Guaranteed Optimal Robust Explanations for NLP Models. In *Proc. Int.*
758 *Joint Conf. on Artificial Intelligence (IJCAI)*, pp. 2658–2665, 2021.

759
760 Eduardo Sany Laber. The Computational Complexity of some Explainable Clustering Problems.
761 *Information Processing Letters*, 184:106437, 2024.

762 Shuai Li, Yingjie Zhang, Hongtu Zhu, Christina Wang, Hai Shu, Ziqi Chen, Zhuoran Sun, and
763 Yanfeng Yang. K-Nearest-Neighbor Local Sampling Based Conditional Independence Testing.
764 *Proc. 37th Conf. on Neural Information Processing Systems (NeurIPS)*, pp. 23321–23344, 2023.

765
766 Gianluigi Lopardo, Frédéric Precioso, and Damien Garreau. A Sea of Words: An In-Depth Analysis
767 of Anchors for Text Data. In *Proc. 26th Int. Conf. on Artificial Intelligence and Statistics (AISTATS)*
768 2023, 2023.

769
770 Scott M Lundberg and Su-In Lee. A Unified Approach to Interpreting Model Predictions. *Proc. 31st*
771 *Conf. on Neural Information Processing Systems (NeurIPS)*, 2017.

772 Joao Marques-Silva and Alexey Ignatiev. Delivering Trustworthy AI through formal XAI. In *Proc.*
773 *36th AAAI Conf. on Artificial Intelligence*, pp. 12342–12350, 2022.

774
775 Joao Marques-Silva and Alexey Ignatiev. No Silver Bullet: Interpretable ML Models Must be
776 Explained. *Frontiers in Artificial Intelligence*, 6:1128212, 2023.

777
778 Joao Marques-Silva, Thomas Gerspacher, Martin Cooper, Alexey Ignatiev, and Nina Narodytska.
779 Explaining Naive Bayes and Other Linear Classifiers with Polynomial Time and Delay. In *Proc.*
780 *33rd Int. Conf. on Advances in Neural Information Processing Systems (NeurIPS)*, pp. 20590–
781 20600, 2020.

782
783 Joao Marques-Silva, Thomas Gerspacher, Martin Cooper, Alexey Ignatiev, and Nina Narodytska.
784 Explanations for Monotonic Classifiers. In *Proc. Int. Conf. on Machine Learning (ICML)*, pp.
785 7469–7479, 2021.

786
787 Reda Marzouk and Colin De La Higuera. On the Tractability of SHAP Explanations under Markovian
788 Distributions. In *Proc. 41st Int. Conf. on Machine Learning (ICML)*, pp. 34961–34986, 2024.

789
790 Reda Marzouk, Shahaf Bassan, and Guy Katz. SHAP Meets Tensor Networks: Provably Tractable
791 Explanations with Parallelism. In *Proc. 39th Conf. on Neural Information Processing Systems*
792 *(NeurIPS)*, 2025a.

793
794 Reda Marzouk, Shahaf Bassan, Guy Katz, and De la Higuera. On the Computational Tractability
795 of the (Many) Shapley Values. In *Proc. 28th Int. Conf. on Artificial Intelligence and Statistics*
796 *(AISTATS)*, 2025b.

797
798 Kuldeep S Meel, Aditya A Shrotri, and Moshe Y Vardi. Not all FPRASs are Equal: Demystifying
799 FPRASs for DNF-Counting. *Constraints*, 24(3):211–233, 2019.

800
801 Ramaravind K Mothilal, Amit Sharma, and Chenhao Tan. Explaining Machine Learning Classifiers
802 Through Diverse Counterfactual Explanations. In *Proc. 3rd Conf. on Fairness, Accountability, and*
803 *Transparency (FAccT)*, pp. 607–617, 2020.

804
805 Vidya Muthukumar, Tejaswini Pedapati, Nalini Ratha, Prasanna Sattigeri, Chai-Wah Wu, Brian
806 Kingsbury, Abhishek Kumar, Samuel Thomas, Aleksandra Mojsilovic, and Kush R Varshney.
807 Understanding Unequal Gender Classification Accuracy from Face Images. 2018. Technical
808 Report. <https://arXiv.org/abs/1812.00099>.

809
810 Sebastian Ordyniak, Giacomo Paesani, and Stefan Szeider. The Parameterized Complexity of Finding
811 Concise Local Explanations. In *Proc. 32nd Int. Joint Conf. on Artificial Intelligence (IJCAI)*, 2023.

812
813 Sebastian Ordyniak, Giacomo Paesani, Mateusz Rychlicki, and Stefan Szeider. Explaining Decisions
814 in ML Models: A Parameterized Complexity Analysis. In *Proc. 21st Int. Conf. on Principles of*
815 *Knowledge Representation and Reasoning (KR)*, pp. 563–573, 2024.

-
- 810 Marco Tulio Ribeiro, Sameer Singh, and Carlos Guestrin. Anchors: High-Precision Model-Agnostic
811 Explanations. In *Proc. 32nd AAAI Conf. on Artificial Intelligence*, 2018.
- 812
- 813 Majun Shi, Zishen Yang, and Wei Wang. Minimum Non-Submodular Cover Problem with Applica-
814 tions. *Applied Mathematics and Computation*, 410:126442, 2021.
- 815
- 816 Andy Shih, Arthur Choi, and Adnan Darwiche. A Symbolic Approach to Explaining Bayesian
817 Network Classifiers. In *Proc. 27th Int. Joint Conf. on Artificial Intelligence (IJCAI)*, pp. 5103–
818 5111, 2018.
- 819
- 820 Bernardo Subercaseaux, Marcelo Arenas, and Kuldeep S Meel. Probabilistic Explanations for Linear
821 Models. In *Proc. 39th AAAI Conf. on Artificial Intelligence*, pp. 20655–20662, 2025.
- 822
- 823 Guy Van den Broeck, Anton Lykov, Maximilian Schleich, and Dan Suciu. On the Tractability of
824 SHAP Explanations. *Journal of Artificial Intelligence Research (JAIR)*, 74:851–886, 2022.
- 825
- 826 Gal Vardi, Gilad Yehudai, and Ohad Shamir. On the Optimal Memorization Power of ReLU Neural
827 Networks. In *Proc. 10th Int. Conf. on Learning Representations, (ICLR)*, 2022.
- 828
- 829 Vijay V Vazirani. *Approximation Algorithms*. Springer, 2001.
- 830
- 831 Sahil Verma, Varich Boonsanong, Minh Hoang, Keegan Hines, John Dickerson, and Chirag Shah.
832 Counterfactual Explanations and Algorithmic Recourses for Machine Learning: A Review. *ACM*
833 *Computing Surveys*, 56(12):1–42, 2024.
- 834
- 835 Stephan Wäldchen, Jan Macdonald, Sascha Hauch, and Gitta Kutyniok. The Computational Com-
836 plexity of Understanding Binary Classifier Decisions. *Journal of Artificial Intelligence Research*
837 *(JAIR)*, 70:351–387, 2021.
- 838
- 839 E. Wang, P. Khosravi, and G. Van den Broeck. Probabilistic Sufficient Explanations. In *Proc. 30th*
840 *Int. Joint Conf. on Artificial Intelligence (IJCAI)*, 2021a.
- 841
- 842 S. Wang, H. Zhang, K. Xu, X. Lin, S. Jana, C. Hsieh, and Z. Kolter. Beta-Crown: Efficient Bound
843 Propagation with Per-Neuron Split Constraints for Neural Network Robustness Verification. In
844 *Proc. 35th Int. Conf. on Advances in Neural Information Processing Systems (NeurIPS)*, pp.
845 29909–29921, 2021b.
- 846
- 847 Greta Warren, Eoin Delaney, Christophe Guéret, and Mark T Keane. Explaining Multiple Instances
848 Counterfactually: User Tests of Group-Counterfactuals for XAI. In *Proc. 32nd Int. Conf. on*
849 *Case-Based Reasoning (ICCBR)*, pp. 206–222, 2024.
- 850
- 851 Ingo Wegener. The Complexity of Symmetric Boolean Functions. *Computation Theory and Logic*,
852 pp. 433–442, 2005.
- 853
- 854 Laurence A Wolsey. An Analysis of the Greedy Algorithm for the Submodular Set Covering Problem.
855 *Combinatorica*, 2(4):385–393, 1982.
- 856
- 857 Haoze Wu, Omri Isac, Aleksandar Zeljić, Teruhiro Tagomori, Matthew Daggitt, Wen Kokke, Idan
858 Refaeli, Guy Amir, Kyle Julian, Shahaf Bassan, et al. Marabou 2.0: A Versatile Formal Analyzer
859 of Neural Networks. In *Proc. 36th Int. Conf. on Computer Aided Verification (CAV)*, pp. 249–264,
860 2024a.
- 861
- 862 Min Wu, Haoze Wu, and Clark Barrett. Verix: Towards Verified Explainability of Deep Neural
863 Networks. In *Proc. 36th Int. Conf. on Advances in Neural Information Processing Systems*
(*NeurIPS*), 2024b.
- 864
- 865 Lei Xu, Maria Skoularidou, Alfredo Cuesta-Infante, and Kalyan Veeramachaneni. Modeling Tabular
866 Data Using Conditional GAN. *Proc. 33rd Cond. on Neural Information Processing Systems*
(*NeurIPS*), 2019.
- 867
- 868 Anton Xue, Rajeev Alur, and Eric Wong. Stability Guarantees for Feature Attributions with Multi-
869 plicative Smoothing. *Proc. 37th Conf. on Neural Information Processing Systems (NeurIPS)*, pp.
870 62388–62413, 2023.

864 Kayo Yin and Graham Neubig. Interpreting Language Models with Contrastive Explanations. In
865 *Proc. Conf. on Empirical Methods in Natural Language Processing (EMNLP)*, pp. 184–198, 2022.
866

867 Jinqiang Yu, Alexey Ignatiev, Peter Stuckey, Nina Narodytska, and Joao Marques-Silva. Eliminating
868 the Impossible, Whatever Remains Must Be True: On Extracting and Applying Background
869 Knowledge in the Context of Formal Explanations. In *Proc. 37th AAAI Conf. on Artificial
870 Intelligence*, pp. 4123–4131, 2023.
871
872
873
874
875
876
877
878
879
880
881
882
883
884
885
886
887
888
889
890
891
892
893
894
895
896
897
898
899
900
901
902
903
904
905
906
907
908
909
910
911
912
913
914
915
916
917

Appendix

The appendix records the formal model and distribution definitions, auxiliary examples, and complete proofs referenced in the main text:

Appendix A provides extended related work on Formal XAI.

Appendix B specifies the model classes and distributional assumptions used in this work.

Appendix C contains the proof of Proposition 1.

Appendix D contains the proof of Proposition 2.

Appendix E contains the proof of Proposition 3.

Appendix F contains the proof of Proposition 4.

Appendix G contains the proof of Proposition 5.

Appendix H contains the proof of Theorem 1.

Appendix I contains the proof of Proposition 6.

Appendix J contains the proof of Theorem 2.

Appendix K contains the proof of Theorem 3.

Appendix L contains the proof of Theorem 4.

Appendix O contains an LLM usage disclosure.

A EXTENDED RELATED WORK ON FORMAL XAI

This work belongs to *formal explainable AI* (formal XAI) (Marques-Silva & Ignatiev, 2022), which studies explanation objects satisfying explicit mathematical guarantees (Yu et al., 2023; Darwiche & Ji, 2022; Darwiche, 2023; Shih et al., 2018; Azzolin et al., 2025; Audemard et al., 2021; Calautti et al., 2025). A primary technical question in this area is the computational complexity of constructing such guaranteed explanations (Barceló et al., 2020; ?; Marzouk et al., 2025a;b; Marzouk & De La Higuera, 2024; Bassan et al., 2025e; Amir et al., 2024; Blanc et al., 2021; 2022; Jaakkola et al., 2025; Amgoud & Cooper, 2026; Geibinger et al., 2025). Because exact construction is frequently intractable for expressive model classes, including neural networks (Adolfi et al., 2024; Barceló et al., 2020) and tree ensembles (Ordyniak et al., 2024; Bassan et al., 2025a), prior work often restricts attention to structurally tractable classes (Marques-Silva & Ignatiev, 2023), such as decision trees (Bounia, 2025; Arenas et al., 2022; Bounia, 2024; Bounia & Koriche, 2023), monotonic models (Marques-Silva et al., 2021; Harzli et al., 2023), linear models (Marques-Silva et al., 2020; Subercaseaux et al., 2025), and additive models (Bassan et al., 2026).

A complementary line of work implements formal explanation computation using automated reasoning and verification technology. Examples include MaxSAT and MILP formulations for tree ensembles (Audemard et al., 2023; 2022a; Ignatiev et al., 2022; Boumazouza et al., 2021), as well as neural-network verification systems (Wang et al., 2021b; Wu et al., 2024a) used to derive certified neural-network explanations (Wu et al., 2024b; Bassan & Katz, 2023; Bassan et al., 2023; Izza et al., 2024a; Hadad et al., 2026; Ignatiev et al., 2019). Other approaches reduce the computational burden by using abstractions (De Palma et al., 2025; Bassan et al., 2025b; Boumazouza et al., 2026), relaxation mechanisms such as smoothing (Xue et al., 2023; Anani et al., 2025; Jin et al., 2025), or training-time interventions (Alvarez Melis & Jaakkola, 2018; Bassan et al., 2025c;d).

Within formal XAI, another related direction applies combinatorial and submodular-optimization techniques to the computation of *local* probabilistic sufficient reasons (Bounia, 2025; Bounia & Koriche, 2023), especially for decision trees. That line of work observes, consistently with our analysis, that the relevant local value functions are not submodular in general. Greedy algorithms inspired by submodular optimization may nevertheless perform well empirically. Our contribution is different: we prove that the *global* value functions themselves satisfy the structural assumptions needed for formal combinatorial-optimization guarantees.

B MODEL AND DISTRIBUTION FORMALIZATIONS

In this section, we formalize the models and distributions referenced throughout the paper. Specifically, Subsection B.1 defines the model families, while Subsection B.2 formalizes the distributions.

B.1 MODEL FORMALIZATIONS

This subsection specifies the base model classes used in the complexity results: (i) (axis-aligned) decision trees, (ii) linear classifiers, and (iii) neural networks with ReLU activations.

Decision Trees. A decision tree (DT) is a directed acyclic graph representing a function $f : \mathbb{F} \rightarrow [c]$, where $c \in \mathbb{N}$ is the number of output classes. The graph representation satisfies: (i) Each internal node v is assigned a distinct binary input feature from the set $\{1, \dots, n\}$; (ii) Every internal node v has at most k outgoing edges, each corresponding to a value in $[k]$ assigned to v ; (iii) Along any path α in the decision tree, each variable appears at most once; (iv) Each leaf node is labeled with a class from $[c]$. An input assignment $\mathbf{x} \in \mathbb{F}$ therefore induces a unique root-to-leaf path α , and the output $f(\mathbf{x})$ is the class label assigned to the reached leaf. The size $|f|$ is the number of graph edges. We allow the variable order to vary across distinct paths α and α' , subject only to the condition that no variable is queried more than once on any single path.

Neural Networks. Our *hardness* proofs are stated for ReLU neural networks, whereas our *tractability results* apply to any architecture with polynomial-time inference. Hence the tractable/intractable separations extend to any neural architecture at least as expressive as a standard feed-forward ReLU network, including common families such as ResNets, CNNs, Transformers, and diffusion models. Any ReLU network can be encoded as a fully connected network by assigning zero weights and biases to missing connections; following standard conventions (Barceló et al., 2020; Bassan et al., 2024; Adolphi et al., 2025), we therefore use fully connected multi-layer perceptrons (MLPs) as the formal representation. An MLP f has $t - 1$ *hidden layers* ($g^{(j)}$ for $j = 1, \dots, t - 1$) and an output layer $g^{(t)}$, with layers defined recursively by:

$$g^{(j)} := \sigma^{(j)}(g^{(j-1)}W^{(j)} + b^{(j)}) \quad (9)$$

Here, $W^{(j)}$ is the weight matrix, $b^{(j)}$ is the bias vector, and $\sigma^{(j)}$ is the activation function in layer j . Thus the model is parameterized by t weight matrices ($W^{(1)}, \dots, W^{(t)}$), t bias vectors ($b^{(1)}, \dots, b^{(t)}$), and t activation functions ($\sigma^{(1)}, \dots, \sigma^{(t)}$).

The input layer is $g^{(0)} = \mathbf{x} \in \{0, 1\}^n$. A sequence of positive integers $d_0, \dots, d_t$ specifies the layer widths, with $W^{(j)} \in \mathbb{Q}^{d_{j-1} \times d_j}$ and $b^{(j)} \in \mathbb{Q}^{d_j}$. These rational parameters are the learned weights and biases. For binary classification over n features, we set $d_0 = n$ and $d_t = 1$. Hidden layers use the ReLU activation $\text{ReLU}(x) = \max(0, x)$. Although sigmoid outputs are common during training, the post-hoc explanation problem can equivalently use a thresholded output layer, defined by $\text{step}(\mathbf{z}) = 1$ if $\mathbf{z} \geq 0$ and $\text{step}(\mathbf{z}) = 0$ otherwise.

Tree Ensembles. Because the objective is post-hoc explanation, the formalization concerns the *inference* map of an ensemble rather than its training procedure. We explicitly consider ensembles that aggregate base-tree predictions by *majority voting* (e.g., Random Forests) or *weighted voting* (e.g., XGBoost). As in the neural-network case, the *tractability* results extend to any ensemble representation whose inference procedure runs in polynomial time.

Majority Voting Inference. Under *majority voting*, the ensemble prediction $f(\mathbf{x})$ is the class $j \in [c]$ that receives the largest number of base-tree predictions $f_i(\mathbf{x})$ over $i \in [k]$.

$$f(\mathbf{x}) := \text{majority}(\{f_i(\mathbf{x}) \mid i \in [k]\}) \quad (10)$$

Here, $\text{majority}(S)$ denotes the most frequent label in the multiset S ; ties are resolved by a fixed deterministic tie-breaking rule, such as lexicographic order or a predefined priority.

Weighted Voting Inference. Under *weighted voting*, base model i is assigned a rational weight $\phi_i \in \mathbb{Q}$, and the ensemble predicts the class with maximum aggregate supporting weight. For any $\mathbf{x} \in \mathbb{F}$, this inference rule is:

$$f(\mathbf{x}) := \arg \max_{j \in [c]} \sum_{i=1}^k \phi_i \cdot \mathbb{I}[f_i(\mathbf{x}) = j] \quad (11)$$

where $\mathbb{I}$ denotes the indicator function.

B.2 DISTRIBUTION FORMALIZATIONS

This subsection specifies the distribution classes referenced in the main text.

Empirical Distributions. An empirical distribution $\mathcal{D}$ is induced by a finite dataset $\mathbf{D}$ containing inputs $\mathbf{z}^1, \mathbf{z}^2, \dots, \mathbf{z}^{|\mathbf{D}|}$. The probability mass assigned to $\mathbf{x} \in \mathbb{F}$ is its relative frequency in $\mathbf{D}$:

$$\Pr(\mathbf{x}) := \frac{1}{|\mathbf{D}|} \sum_{i=1}^{|\mathbf{D}|} \mathbb{I}(\mathbf{z}^i = \mathbf{x}) \quad (12)$$

Independent Distributions. Given per-coordinate probabilities $p(\mathbf{x}_i) \in [0, 1]$, the distribution $\mathcal{D}$ is independent when its joint probability factorizes as $\Pr(\mathbf{x}) := \prod_{i \in [n]} p(\mathbf{x}_i)$.

When $p(\mathbf{x}_i) = p(\mathbf{x}_j)$ for all $i, j \in [n]$, this factorized distribution specializes to the *uniform distribution*.

General Distributions. Several proofs, especially those establishing structural properties of value functions, hold without imposing a special representation on $\mathcal{D}$. Empirical distributions are common in XAI because they use the available dataset as a proxy for the data-generating process, but other distributional approximations include k-NN resampling from nearby points (Li et al., 2023; Almanjahie et al., 2018; Gweon et al., 2019), copulas for tabular dependence modeling (Größer & Okhrin, 2022), CTGAN (Xu et al., 2019), and conditional diffusion models (Huang et al., 2022). Such distributional choices are also common in counterfactual explanation and algorithmic recourse (Karimi et al., 2021; Fokkema et al., 2024; 2023; Verma et al., 2024), which are closely related to contrastive explanations. When v cannot be evaluated exactly from model structure (for example, by leaf enumeration in decision trees) or from an empirical distribution, it may instead be estimated by Monte Carlo sampling (Hastings, 1970), as in related probabilistic-explanation approaches (Subercaseaux et al., 2025; Lopardo et al., 2023).

B.3 SUBSET VS. CARDINAL MINIMALITY

This subsection makes explicit the logical relationship between cardinal minimality and subset minimality. Cardinal minimality is the stronger requirement because it minimizes explanation size globally, whereas subset minimality only excludes qualifying proper subsets.

Indeed, if $S \subseteq [n]$ is cardinally minimal, then no explanation S' with $|S'| < |S|$ exists. Every strict subset $S' \subsetneq S$ has smaller cardinality, and therefore cannot be an explanation; hence S is subset-minimal.

The converse implication fails. Consider:

$$f := \mathbf{x}_1 \vee (\mathbf{x}_2 \wedge \mathbf{x}_3) \quad (13)$$

with assignment $\mathbf{x} := (1, 1, 1)$, so that $f(1, 1, 1) = 1$. Fixing $\mathbf{x}_1 = 1$ gives a cardinally minimal, and hence subset-minimal, sufficient reason because the output remains 1 for all assignments of $\mathbf{x}_2$ and $\mathbf{x}_3$. Fixing both $\mathbf{x}_2 = 1$ and $\mathbf{x}_3 = 1$ also gives a subset-minimal sufficient reason, but it is not cardinally minimal. Thus every cardinally minimal explanation is subset-minimal, while some subset-minimal explanations are not cardinally minimal.

C PROOF OF PROPOSITION 1

Proposition 1. *While the non-probabilistic case ($\delta = 1$) admits a unique subset-minimal global sufficient reason (Bassan et al., 2024), in the general probabilistic setting (for arbitrary δ), there exist functions f that have $\Theta(\frac{2^n}{\sqrt{n}})$ subset-minimal global sufficient reasons.*

Proof. It is known that certain Boolean functions admit an exponential number of subset-minimal local (non-probabilistic) sufficient reasons for some input $\mathbf{x}$ Bassan et al. (2024). In particular, this phenomenon was shown to occur in functions of the following form. We demonstrate that the same

function admits an exponential number of *global* and *probabilistic* sufficient reasons. Notably, the mentioned function is a special case of a broader class of *threshold Boolean functions* described in Wegener (2005). Specifically, for $n = 2k + 1$ with $k \in \mathbb{N}$, this function is defined as:

$$f(\mathbf{x}) := \begin{cases} 1 & \text{if } \sum_{i=1}^n \mathbf{x}_i \geq k + 1 \\ 0 & \text{otherwise} \end{cases} \quad (14)$$

where $\mathbf{x}$ is drawn from a uniform distribution $\mathcal{D}$. Notably, the function f is *symmetric*, in the sense of symmetric threshold Boolean functions Wegener (2005): its output depends solely on the number of 1's (or, equivalently, 0's) in the input and is invariant under any permutation of input bits. Furthermore, the two extreme inputs — the all-zeros vector $(0, \dots, 0)$ and the all-ones vector $(1, \dots, 1)$ — each admit an exponential number of subset-minimal local (non-probabilistic) sufficient reasons.

Since the function f is symmetric, the condition $v_{\text{suff}}^g(S_1) = v_{\text{suff}}^g(S_2)$ holds for any two subsets $S_1, S_2 \subseteq [n]$. Moreover, it can be verified that for any subset $S \subseteq [n]$ of size $k + 1$, we have:

$$(\forall i \in S). \quad v_{\text{suff}}^g(S) > v_{\text{suff}}^g(S \setminus \{i\}) \quad (15)$$

We also deliberately set δ to satisfy:

$$v_{\text{suff}}^g(S \setminus \{i\}) < \delta \leq v_{\text{suff}}^g(S), \quad (16)$$

and also set $k := \lfloor \frac{n}{2} \rfloor$. Each of these subsets is *subset-minimal* because any subset of size at most $\lfloor \frac{n}{2} \rfloor - 1$ fails to be a sufficient reason for $\langle f, \mathbf{x} \rangle$ (with respect to the δ threshold). Therefore, we can directly apply the same analysis as in (Bassan et al., 2024). In particular, there are exactly $\binom{n}{\lfloor \frac{n}{2} \rfloor}$ subset-minimal local sufficient reasons for $\langle f, \mathbf{x} \rangle$. Using Stirling's approximation, we obtain:

$$\lim_{n \rightarrow \infty} \frac{2\sqrt{2\pi}}{e^2} \cdot \frac{2^n}{\sqrt{n}} \leq \binom{n}{\lfloor \frac{n}{2} \rfloor} \leq \frac{e}{\pi} \cdot \frac{2^n}{\sqrt{n}} \quad (17)$$

This yields the corresponding bound on the number of subset-minimal global δ -sufficient reasons.

□

D PROOF OF PROPOSITION 2

Proposition 2. *While the local probabilistic setting (for any δ) lacks monotonicity — i.e., the value functions v_{con}^ℓ and v_{suff}^ℓ are non-monotonic (Arenas et al., 2022; Izza et al., 2023; Subercaseaux et al., 2025; Izza et al., 2024b) — in the global probabilistic setting (also for any δ), both value functions v_{con}^g and v_{suff}^g are monotonic non-decreasing.*

Proof. In this section, we prove the monotonicity property for both types of global explanations — sufficient and contrastive. We begin by establishing the property for global sufficiency. To build intuition, we first focus on the simpler case of Boolean functions, then generalize the result to functions with discrete multi-valued input domains and multiple output classes. We further extend the result to continuous input domains and show that the monotonicity property holds for any well-defined classification function. Finally, we demonstrate how the same monotonicity guarantees apply to global contrastive explanations. Importantly, none of our proofs rely on the assumption of feature independence — the monotonicity property holds strongly for *any* underlying distribution.

Lastly, we present a simple example showing that the monotonicity property does not hold for the local probabilistic sufficient or contrastive value functions.

Lemma 1. *The global sufficient value function v_{suff}^g is monotonic non-decreasing for Boolean functions, under any data distribution.*

Proof. Given an arbitrary set $S \in [n]$, and a feature $i \notin S$, we have that:

$$\Pr(\mathbf{x}) = \Pr(\mathbf{x}_S, \mathbf{x}_{\bar{S}}) = \Pr(\mathbf{x}_{\bar{S}} | \mathbf{x}_S) \cdot \Pr(\mathbf{x}_S) \quad (18)$$

Consequently, this implies that:

$$\begin{aligned} \Pr(\mathbf{x}) &= \Pr(\mathbf{x}_{\overline{S \cup \{i\}}} | \mathbf{x}_{S \cup \{i\}}) \cdot \Pr(\mathbf{x}_{S \cup \{i\}}) \\ &= \Pr(\mathbf{x}_{\overline{S \cup \{i\}}} | \mathbf{x}_{S \cup \{i\}}) \cdot \Pr(\mathbf{x}_S, x_i) \\ &= \Pr(\mathbf{x}_{\overline{S \cup \{i\}}} | \mathbf{x}_{S \cup \{i\}}) \cdot \Pr(x_i | \mathbf{x}_S) \cdot \Pr(\mathbf{x}_S) \end{aligned} \quad (19)$$

Moreover, for any two points $\mathbf{x}, \mathbf{x}' \sim \mathcal{D}$ such that $\mathbf{x}_S = \mathbf{x}'_S$ and $f(\mathbf{x}) = 1 - f(\mathbf{x}')$, it holds that:

$$\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) | \mathbf{z}_S = \mathbf{x}_S) = 1 - \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}') | \mathbf{z}_S = \mathbf{x}'_S) \quad (20)$$

To simplify notation, we occasionally omit the distribution notation $\sim \mathcal{D}$ in the local probability expression $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) | \mathbf{z}_S = \mathbf{x}_S)$, and use f^+ and f^- to denote the events $f(\mathbf{z}) = 1$ and $f(\mathbf{z}) = 0$, respectively.

We begin with a technical simplification of the probability term to facilitate the proof, and then proceed to establish monotonicity.

Simplifying the probability term. Let $\mathcal{D}_S$ denote the distribution restricted to the set S . Given a point $\mathbf{x} \sim \mathcal{D}$ and a set S , consider all points $\mathbf{z} \sim \mathcal{D}$ such that $\mathbf{z}_S = \mathbf{x}_S$ and $f(\mathbf{z}) = b$, where $b \in \{0, 1\}$ — they all share the same local probability $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = b | \mathbf{z}_S = \mathbf{x}_S)$. For every $\mathbf{x} \sim \mathcal{D}$ that shares the same x_S and the same output b , we marginalize over $x_{\bar{S}}$. This yields $\Pr(f(\mathbf{x}) = b, \mathbf{x}_S) = \Pr(f(\mathbf{x}) = b | \mathbf{x}_S) \cdot \Pr(\mathbf{x}_S)$, from which we can infer:

$$\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = b | \mathbf{z}_S = \mathbf{x}_S) = \Pr(f(\mathbf{x}) = b | \mathbf{x}_S). \quad (21)$$

Thus, for $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) | \mathbf{z}_S = \mathbf{x}_S)]$, we get that:

$$\begin{aligned} &\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) | \mathbf{z}_S = \mathbf{x}_S)] \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S, f^+} \Pr(f^+, \mathbf{x}_S) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f^+ | \mathbf{z}_S = \mathbf{x}_S) + \sum_{\mathbf{x}_S \sim \mathcal{D}_S, f^-} \Pr(f^-, \mathbf{x}_S) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f^- | \mathbf{z}_S = \mathbf{x}_S) \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [(\Pr_{\mathbf{z} \sim \mathcal{D}}(f^+ | \mathbf{z}_S = \mathbf{x}_S))^2 + (\Pr_{\mathbf{z} \sim \mathcal{D}}(f^- | \mathbf{z}_S = \mathbf{x}_S))^2] \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [(\Pr_{\mathbf{z} \sim \mathcal{D}}(f^+ | \mathbf{z}_S = \mathbf{x}_S))^2 + (1 - \Pr_{\mathbf{z} \sim \mathcal{D}}(f^+ | \mathbf{z}_S = \mathbf{x}_S))^2], \end{aligned} \quad (22)$$

From which it follows that:

$$\begin{aligned} &\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) | \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})] \\ &= \sum_{\mathbf{x}_{S \cup \{t\}} \sim \mathcal{D}_{S \cup \{t\}}} \Pr(\mathbf{x}_{S \cup \{t\}}) \cdot [(\Pr_{\mathbf{z}}(f^+ | \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}}))^2 + (1 - \Pr_{\mathbf{z}}(f^+ | \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}}))^2] \\ &= \sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_t=1}} \Pr(x_t = 1, \mathbf{x}_S) \cdot [(\Pr_{\mathbf{z}}(f^+ | \mathbf{z}_S = \mathbf{x}_S, z_t = 1))^2 + (1 - \Pr_{\mathbf{z}}(f^+ | \mathbf{z}_S = \mathbf{x}_S, z_t = 1))^2] + \\ &\quad \sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_t=0}} \Pr(x_t = 0, \mathbf{x}_S) \cdot [(\Pr_{\mathbf{z}}(f^+ | \mathbf{z}_S = \mathbf{x}_S, z_t = 0))^2 + (1 - \Pr_{\mathbf{z}}(f^+ | \mathbf{z}_S = \mathbf{x}_S, z_t = 0))^2] \end{aligned} \quad (23)$$

Proving monotonicity. We now proceed to prove the monotonicity claim, building on the previous simplification. We begin by introducing a few additional notations. Let $g_1^+ := \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 | \mathbf{z}_S = \mathbf{x}_S, z_t = 1)$ and $g_0^+ := \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 | \mathbf{z}_S = \mathbf{x}_S, z_t = 0)$. Similarly, define $P_1 := \Pr(z_t =$

1 | $\mathbf{x}_S$) and $P_0 := \Pr(z_t = 0 | \mathbf{x}_S)$. Then, the following expectations $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) | \mathbf{z}_S = \mathbf{x}_S)]$ and $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) | \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})]$ can be simplified accordingly¹:

$$\begin{aligned} & \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) | \mathbf{z}_S = \mathbf{x}_S)] \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [(P_1 \cdot g_1^+ + P_0 \cdot g_0^+)^2 + (1 - (P_1 \cdot g_1^+ + P_0 \cdot g_0^+))^2] \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [2(P_1 \cdot g_1^+ + P_0 \cdot g_0^+)^2 - 2(P_1 \cdot g_1^+ + P_0 \cdot g_0^+) + 1]. \end{aligned} \quad (24)$$

As a result, we obtain the following:

$$\begin{aligned} & \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) | \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})] \\ &= \sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_t=1}} \Pr(\mathbf{x}_S) \cdot P_1 \cdot ((g_1^+)^2 + (1 - g_1^+)^2) + \sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_t=0}} \Pr(\mathbf{x}_S) \cdot P_0 \cdot ((g_0^+)^2 + (1 - g_0^+)^2) \\ &= \sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_t=1}} \Pr(\mathbf{x}_S) \cdot P_1 \cdot (2(g_1^+)^2 - 2g_1^+ + 1) + \sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_t=0}} \Pr(\mathbf{x}_S) \cdot P_0 \cdot (2(g_0^+)^2 - 2g_0^+ + 1) \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [P_1 \cdot (2(g_1^+)^2 - 2g_1^+ + 1) + P_0 \cdot (2(g_0^+)^2 - 2g_0^+ + 1)] \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [2P_1 \cdot ((g_1^+)^2 - g_1^+) + 2P_0 \cdot ((g_0^+)^2 - g_0^+) + 1] \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [2(P_1 \cdot (g_1^+)^2 + P_0 \cdot (g_0^+)^2) - 2(P_1 \cdot g_1^+ + P_0 \cdot g_0^+) + 1]. \end{aligned} \quad (25)$$

Let $\Delta_t(S, f)$ denote $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) | \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})] - \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) | \mathbf{z}_S = \mathbf{x}_S)]$. Without loss of generality, we assume that both P_1 and P_0 are positive. Otherwise, it can be verified that $\Delta_t(S, f) = 0$. Then, we get:

$$\begin{aligned} & \Delta_t(S, f) \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [2(P_1 \cdot (g_1^+)^2 + P_0 \cdot (g_0^+)^2) - 2(P_1 \cdot g_1^+ + P_0 \cdot g_0^+)] \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} 2 \cdot \Pr(\mathbf{x}_S) \cdot [(P_1 - P_1^2)(g_1^+)^2 + (P_0 - P_0^2)(g_0^+)^2 - 2 \cdot P_1 \cdot P_0 \cdot g_1^+ \cdot g_0^+] \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} 2 \cdot \Pr(\mathbf{x}_S) \cdot P_1 \cdot P_0 \cdot (g_1^+ - g_0^+)^2 \\ &\geq 0. \end{aligned} \quad (26)$$

Note that $P_1 - P_1^2 = P_1(1 - P_1) = P_1 P_0$ and similarly: $P_0 - P_0^2 = P_0(1 - P_0) = P_0 P_1$. Using these identities along with the previous result, we conclude that $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) | \mathbf{z}_S = \mathbf{x}_S)]$ is monotone, thereby concluding our proof. $\square$

Lemma 2. *The global sufficient value function v_{uff}^g is monotonic non-decreasing for functions with discrete multi-valued input domains and multiple output classes, under any data distribution.*

Proof. Extending the proof from the simplified Boolean case, we now generalize the result to functions whose input domains and output ranges each consist of a finite set of values. Let $\mathcal{K}$ represent the set of output classes.

¹Note that the notations P_1 , P_0 , g_1^+ , and g_0^+ —as well as all others used throughout the paper—are not fixed constants, but rather depend on the specific input $\mathbf{x}_S$. Formally, they should be written as $P_1(\mathbf{x}_S)$, $P_0(\mathbf{x}_S)$, and so on. However, for readability, we omit the explicit dependence on $\mathbf{x}_S$.

We can generalize equation 22 as follows:

$$\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] = \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} (\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S))^2 \right] \quad (27)$$

Given the condition $\mathbf{z}_S = \mathbf{x}_S$, suppose feature t can take r possible values $\{v_1, \dots, v_r\}$. Then, equation 23 can be generalized as follows:

$$\begin{aligned} & \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})] \\ &= \sum_{k=1}^r \left[\sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_t = v_k}} \Pr(\mathbf{x}_S) \cdot \Pr(z_t = v_k \mid \mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} (\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S, z_t = v_k))^2 \right] \right]. \end{aligned} \quad (28)$$

Let P_k denote $\Pr(z_t = v_k \mid \mathbf{x}_S)$, Let g_k^j denote $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S, z_t = v_k)$, that is, z_t takes the value v_k and the output class is j . Then, we have

$$\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] = \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \left(\sum_{k=1}^r P_k \cdot g_k^j \right)^2 \right] \quad (29)$$

and

$$\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})] = \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \left[\sum_{k=1}^r P_k \cdot \left(\sum_{j \in \mathcal{K}} (g_k^j)^2 \right) \right]$$

Combining these two implications, we can conclude that:

$$\begin{aligned} & \Delta_t(S, f) \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \left[\sum_{k=1}^r P_k \cdot \left(\sum_{j \in \mathcal{K}} (g_k^j)^2 \right) - \sum_{j \in \mathcal{K}} \left(\sum_{k=1}^r P_k \cdot g_k^j \right)^2 \right] \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \left(\sum_{k=1}^r P_k \cdot (g_k^j)^2 \right) - \sum_{j \in \mathcal{K}} \left(\sum_{k=1}^r (P_k \cdot g_k^j)^2 + \sum_{k < l} 2 \cdot P_k \cdot P_l \cdot g_k^j \cdot g_l^j \right) \right] \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \sum_{k \neq l} P_k \cdot P_l \cdot (g_k^j)^2 - \sum_{j \in \mathcal{K}} \sum_{k < l} 2 \cdot P_k \cdot P_l \cdot g_k^j \cdot g_l^j \right] \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \sum_{j \in \mathcal{K}} \left[\sum_{k < l} P_k \cdot P_l \cdot (g_k^j - g_l^j)^2 \right] \\ &\geq 0. \end{aligned} \quad (30)$$

Hence, $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)]$ is monotone, which completes the proof for this setting. $\square$

Lemma 3. *The global sufficient value function v_{suff}^g is monotonic non-decreasing for functions with continuous input domains and multiple output classes, under any data distribution.*

Proof. We extend the monotonicity results to functions where features are defined over continuous domains. One approach is to partition the domain of the continuous features into two disjoint subdomains and then adapt the proofs in lemma 2 by revising the notation accordingly. The core reasoning remains unchanged when transitioning from the discrete to the continuous case—only the notation differs.

Alternatively, we provide a direct proof below. Assume all features lie in continuous domains, and let $p(\mathbf{x}_S, \mathbf{x}_{\bar{S}})$ denote the corresponding joint probability density function. Moreover, let $p(x_i)$ denote the marginal probability density function of x_i , and $\mathcal{X}_i$ denote the domain of feature i .

We define marginal density as

$$p(\mathbf{x}_S) := \int_{\mathbf{x}_{\bar{S}} \in \mathcal{X}_{\bar{S}}} p(\mathbf{x}_S, \mathbf{x}_{\bar{S}}) d\mathbf{x}_{\bar{S}}, \quad (31)$$

where $\mathcal{X}_{\bar{S}}$ represents the domain of features in $\bar{S}$.

And the conditional density is defined as:

$$p(x_t | \mathbf{x}_S) := \frac{p(\mathbf{x}_S, x_t)}{p(\mathbf{x}_S)}. \quad (32)$$

The local probability of S given $\mathbf{x}$ is defined as

$$\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j | \mathbf{z}_S = \mathbf{x}_S) = \frac{\int_{\mathbf{z}_{\bar{S}} \in \mathcal{X}_{\bar{S}}} I(f(\mathbf{x}_S, \mathbf{z}_{\bar{S}}) = j) \cdot p(\mathbf{x}_S, \mathbf{z}_{\bar{S}}) d\mathbf{z}_{\bar{S}}}{p(\mathbf{x}_S)}, \quad (33)$$

This term can also be decomposed into:

$$\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j | \mathbf{z}_S = \mathbf{x}_S) = \int_{z_t \in \mathcal{X}_t} p(z_t | \mathbf{x}_S) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j | \mathbf{z}_S = \mathbf{x}_S, z_t = x_t) dz_t \quad (34)$$

via an additional feature $t \notin S$.

We then modify equation 27 to

$$\begin{aligned} & \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) | \mathbf{z}_S = \mathbf{x}_S)] \\ &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} (\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j | \mathbf{z}_S = \mathbf{x}_S))^2 \right] d\mathbf{x}_S, \end{aligned} \quad (35)$$

while equation 28 is changed to

$$\begin{aligned} & \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) | \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})] \\ &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\int_{z_t \in \mathcal{X}_t} p(z_t | \mathbf{x}_S) \cdot \left(\sum_{j \in \mathcal{K}} (\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j | \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}}))^2 \right) dz_t \right] d\mathbf{x}_S. \end{aligned} \quad (36)$$

For simplicity, let $Y_j(t)$ denote $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j | \mathbf{z}_S = \mathbf{x}_S, z_t = v_t)$, and $p_{t|S}$ denote $p(z_t | \mathbf{x}_S)$. Note that both $Y_j(t)$ and $p_{t|S}$ are associated with a fixed $\mathbf{x}_S$ ². We prove monotonicity as follows:

$$\begin{aligned} & \Delta_t(S, f) \\ &= \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) | \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})] - \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) | \mathbf{z}_S = \mathbf{x}_S)] \\ &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\int_{z_t \in \mathcal{X}_t} p_{t|S} \cdot \left(\sum_{j \in \mathcal{K}} (\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j | \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}}))^2 \right) dz_t \right] d\mathbf{x}_S - \\ & \quad \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \left(\int_{z_t \in \mathcal{X}_t} p_{t|S} \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j | \mathbf{z}_S = \mathbf{x}_S, z_t = x_t) dz_t \right)^2 \right] d\mathbf{x}_S \\ &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \left(\int_{\mathcal{X}_t} p_{t|S} \cdot Y_j(t)^2 dz_t - \left(\int_{\mathcal{X}_t} p_{t|S} \cdot Y_j(t) dz_t \right)^2 \right) \right] d\mathbf{x}_S \\ &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \frac{2}{2} \cdot \left(\int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_{t|S} \cdot p_{t'|S} \cdot (Y_j(t)^2 - Y_j(t)Y_j(t')) dz_t dz_{t'} \right) \right] d\mathbf{x}_S \\ &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \frac{1}{2} \cdot \left(\int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_{t|S} \cdot p_{t'|S} \cdot (Y_j(t) - Y_j(t'))^2 dz_t dz_{t'} \right) \right] d\mathbf{x}_S \\ &\geq 0. \end{aligned} \quad (37)$$

²Changing $\mathbf{x}_S$ alters their values, we omit $\mathbf{x}_S$ in the notation for brevity.

1350

1351

1352

1353

1354

1355

1356

1357

1358

1359

1360

1361

1362

1363

1364

1365

1366

1367

1368

1369

1370

1371

1372

1373

1374

1375

1376

1377

1378

1379

1380

1381

1382

1383

1384

1385

1386

1387

1388

1389

1390

1391

1392

1393

1394

1395

1396

1397

1398

1399

1400

1401

1402

1403

□

Proposition 2. *While the local probabilistic setting (for any δ) lacks monotonicity — i.e., the value functions v_{con}^ℓ and v_{suff}^ℓ are non-monotonic (Arenas et al., 2022; Izza et al., 2023; Subercaseaux et al., 2025; Izza et al., 2024b) — in the global probabilistic setting (also for any δ), both value functions v_{con}^g and v_{suff}^g are monotonic non-decreasing.*

Proof. As a consequence of lemmas 1 to 3, we conclude that the global sufficient value function v_{suff}^g is monotonic non-decreasing for any well-defined classification function — whether its features are Boolean, discrete, continuous, or a combination — under any data distribution.

$$v_{\text{con}}^g(S) = 1 - v_{\text{suff}}^g(\bar{S}) \quad (38)$$

This implies that v_{con}^g is monotonic non-increasing with respect to adding features to $\bar{S}$. Equivalently, v_{con}^g is monotonic non-decreasing as features are removed from $\bar{S}$ and added to S , thus completing the proof.

□

E PROOF OF PROPOSITION 3

Proposition 3. *While the local probabilistic sufficient setting (for any δ) lacks supermodularity — even when $\mathcal{D}$ is uniform, i.e., the value function v_{suff}^ℓ is not supermodular — in the global probabilistic setting (also for any δ), when $\mathcal{D}$ exhibits feature independence, the value function v_{suff}^g is supermodular.*

Proof. This section establishes the supermodularity property for both global sufficient and contrastive explanations. As with monotonicity, we begin with Boolean functions and progressively generalize to multi-valued discrete and continuous input domains, and finally consider the case of well-defined classification functions. Unlike the monotonicity property, however, these proofs require the assumption of feature independence. We also provide a counterexample to show that supermodularity may fail when this assumption is relaxed.

Lemma 4. *The global sufficient value function v_{suff}^g is supermodular for Boolean functions, under the assumption of feature independence.*

Proof. Particularly, we will prove that:

$$\Delta_t(S, f) \leq \Delta_t(S', f), \quad (39)$$

where $S \subseteq S'$ and $t \notin S'$. That is, $\Delta_t(S, f)$ is monotone.

Clearly, if $S' = S$, then $\Delta_t(S, f) \leq \Delta_t(S', f)$ holds. Let $S' = S \cup \{i\}$, where $i \notin S$ and $i \neq t$. For the local probability $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)$, under the assumption of feature independence, we have that the following holds:

$$\begin{aligned} \Pr_{\mathbf{z}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S) &= \Pr(z_i = 1) \cdot \Pr(z_t = 1) \cdot \Pr_{\mathbf{z}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 1, z_t = 1) + \\ &\quad \Pr(z_i = 1) \cdot \Pr(z_t = 0) \cdot \Pr_{\mathbf{z}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 1, z_t = 0) + \\ &\quad \Pr(z_i = 0) \cdot \Pr(z_t = 1) \cdot \Pr_{\mathbf{z}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 0, z_t = 1) + \\ &\quad \Pr(z_i = 0) \cdot \Pr(z_t = 0) \cdot \Pr_{\mathbf{z}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 0, z_t = 0). \end{aligned} \quad (40)$$

Let P_1, P_0, Q_1 , and Q_0 denote $\Pr(z_t = 1), \Pr(z_t = 0), \Pr(z_i = 1)$, and $\Pr(z_i = 0)$, respectively. Moreover, let $g_{11}^+, g_{10}^+, g_{01}^+$, and g_{00}^+ denote $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 1, z_t = 1), \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 1, z_t = 0), \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 0, z_t = 1)$, and $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 0, z_t = 0)$, respectively.

We further decompose equation 26 into:

$$\begin{aligned}
\Delta_t(S, f) &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} 2 \cdot \Pr(\mathbf{x}_S) \cdot P_1 P_0 \cdot [(Q_1 \cdot g_{11}^+ + Q_0 \cdot g_{01}^+) - (Q_1 \cdot g_{10}^+ + Q_0 \cdot g_{00}^+)]^2 \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} 2 \cdot \Pr(\mathbf{x}_S) \cdot P_1 P_0 \cdot [Q_1 \cdot (g_{11}^+ - g_{10}^+) + Q_0 \cdot (g_{01}^+ - g_{00}^+)]^2.
\end{aligned} \tag{41}$$

Similarly, the following also holds:

$$\Delta_t(S \cup \{i\}, f) = \sum_{\mathbf{x}_S \sim \mathcal{D}_S} 2 \cdot \Pr(\mathbf{x}_S) \cdot P_1 P_0 \cdot [Q_1 \cdot (g_{11}^+ - g_{10}^+)^2 + Q_0 \cdot (g_{01}^+ - g_{00}^+)^2]. \tag{42}$$

Thus, we get

$$\begin{aligned}
\Delta_t(S \cup \{i\}, f) - \Delta_t(S, f) &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} 2 \cdot \Pr(\mathbf{x}_S) \cdot P_1 P_0 \cdot Q_1 Q_0 \cdot (g_{11}^+ - g_{10}^+ - g_{01}^+ + g_{00}^+)^2 \\
&\geq 0.
\end{aligned} \tag{43}$$

This implies that $\Delta_t(S, f)$ is monotone, that is, $c(S \cup \{t\}, f) - c(S, f) \leq c(S' \cup \{t\}, f) - c(S', f)$. $\square$

Lemma 5. *The global sufficient value function v_{suff}^g is supermodular for functions with discrete multi-valued input domains and multiple output classes, under the assumption of feature independence.*

Proof. We extend the results of the previous section to multi-valued discrete functions, again with the assumption of feature independence. We use P_k to denote $\Pr(z_t = v_k)$, and $Q_{k'}$ to denote $\Pr(z_i = v'_{k'})$. Moreover, we use $g_{k'k}^j$ to denote $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S, z_i = v'_{k'}, z_t = v_k)$. We decompose equation 30 to

$$\begin{aligned}
\Delta_t(S, f) &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \sum_{j \in \mathcal{K}} \left[\sum_{k < l} P_k P_l \cdot \left(\sum_{k'=1}^{r'} Q_{k'} \cdot g_{k'k}^j - \sum_{k'=1}^{r'} Q_{k'} \cdot g_{k'l}^j \right)^2 \right] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \sum_{j \in \mathcal{K}} \left[\sum_{k < l} P_k P_l \cdot \left(\sum_{k'=1}^{r'} Q_{k'} \cdot (g_{k'k}^j - g_{k'l}^j) \right)^2 \right].
\end{aligned} \tag{44}$$

Likewise, we have

$$\begin{aligned}
\Delta_t(S \cup \{i\}, f) &= \sum_{k'=1}^{r'} \left[\sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_i = v'_{k'}}} \Pr(\mathbf{x}_S) \cdot \Pr(z_i = v'_{k'}) \cdot \sum_{j \in \mathcal{K}} \left(\sum_{k < l} P_k P_l \cdot (g_{k'k}^j - g_{k'l}^j)^2 \right) \right] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \left[\sum_{k'=1}^{r'} Q_{k'} \cdot \sum_{j \in \mathcal{K}} \left(\sum_{k < l} P_k P_l \cdot (g_{k'k}^j - g_{k'l}^j)^2 \right) \right] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \sum_{j \in \mathcal{K}} \left[\sum_{k < l} P_k P_l \cdot \sum_{k'=1}^{r'} Q_{k'} \cdot (g_{k'k}^j - g_{k'l}^j)^2 \right].
\end{aligned} \tag{45}$$

Therefore, we get

$$\begin{aligned}
&\Delta_t(S \cup \{i\}, f) - \Delta_t(S, f) \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \sum_{j \in \mathcal{K}} \left[\sum_{k < l} \sum_{k' < l'} P_k P_l Q_{k'} Q_{l'} \cdot (g_{k'k}^j - g_{k'l}^j - g_{l'k}^j + g_{l'l}^j)^2 \right] \\
&\geq 0.
\end{aligned} \tag{46}$$

$\square$

Proposition 3. *While the local probabilistic sufficient setting (for any δ) lacks supermodularity — even when $\mathcal{D}$ is uniform, i.e., the value function v_{suff}^ℓ is not supermodular — in the global probabilistic setting (also for any δ), when $\mathcal{D}$ exhibits feature independence, the value function v_{suff}^g is supermodular.*

Proof. We can generalize the supermodularity results established in Lemma 5 to the continuous case, assuming that $\mathcal{D}_p$ satisfies feature independence. One approach is to partition the domain of the continuous features into two disjoint subdomains and then adapt the proofs in Lemma 5 by revising the notation accordingly.

Alternatively, we provide a direct proof below. Under this assumption, the marginal and conditional densities simplify as follows:

$$p(\mathbf{x}_S) = \prod_{i \in S} p(x_i), \text{ and } p(x_t | \mathbf{x}_S) = p(x_t). \quad (47)$$

Consequently, the local probability of S conditioned on $\mathbf{x}$ becomes:

$$\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j | \mathbf{z}_S = \mathbf{x}_S) = \int_{\mathbf{z}_{\bar{S}} \in \mathcal{X}_{\bar{S}}} I(f(\mathbf{x}_S, \mathbf{z}_{\bar{S}}) = j) \cdot \prod_{i \in \bar{S}} p(z_i) d\mathbf{z}_{\bar{S}}. \quad (48)$$

We let $Y_j(i, t)$ denote $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j | \mathbf{z}_S = \mathbf{x}_S, z_i = x_i, z_t = x_t)$, where $i \notin S, t \notin S$ and $i \neq t$. Evidently,

$$Y_j(i, t) = Y_j(t, i), \text{ and } Y_j(t) = \int_{\mathcal{X}_i} p_i \cdot Y_j(i, t) dz_i.$$

We have:

$$\begin{aligned} \Delta_t(S, f) &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \frac{1}{2} \left(\int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_t p_{t'} (Y_j(t) - Y_j(t'))^2 dz_t dz_{t'} \right) \right] d\mathbf{x}_S \\ &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \frac{1}{2} \left(\int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_t p_{t'} \left(\int_{\mathcal{X}_i} p_i \cdot Y_j(i, t) dz_i - \int_{\mathcal{X}_i} p_i \cdot Y_j(i, t') dz_i \right)^2 dz_t dz_{t'} \right) \right] d\mathbf{x}_S \\ &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \frac{1}{2} \left(\int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_t p_{t'} \left(\int_{\mathcal{X}_i} p_i \cdot (Y_j(i, t) - Y_j(i, t')) dz_i \right)^2 dz_t dz_{t'} \right) \right] d\mathbf{x}_S, \end{aligned} \quad (49)$$

where

$$\begin{aligned} &\int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_t p_{t'} \left(\int_{\mathcal{X}_i} p_i \cdot (Y_j(i, t) - Y_j(i, t')) dz_i \right)^2 dz_t dz_{t'} \\ &= \int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} \int_{\mathcal{X}_i} \int_{\mathcal{X}_{i'}} p_t p_{t'} p_i p_{i'} \cdot (Y_j(i, t) - Y_j(i, t')) \cdot (Y_j(i', t) - Y_j(i', t')) dz_i dz_{i'} dz_t dz_{t'}. \end{aligned}$$

Let $S' = S \cup \{i\}$, then we have:

$$\begin{aligned} \Delta_t(S', f) &= \int_{\mathbf{x}_{S'} \in \mathcal{X}_{S'}} p(\mathbf{x}_{S'}) \cdot \left[\sum_{j \in \mathcal{K}} \frac{1}{2} \left(\int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_t p_{t'} \cdot (Y_j(i, t) - Y_j(i, t'))^2 dz_t dz_{t'} \right) \right] d\mathbf{x}_{S'} \\ &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \frac{1}{2} \int_{\mathcal{X}_i} p_i \cdot \left(\int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_t p_{t'} \cdot (Y_j(i, t) - Y_j(i, t'))^2 dz_t dz_{t'} \right) dz_i \right] d\mathbf{x}_S \quad (50) \\ &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \frac{1}{2} \left(\int_{\mathcal{X}_i} \int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_i p_t p_{t'} \cdot (Y_j(i, t) - Y_j(i, t'))^2 dz_t dz_{t'} dz_i \right) \right] d\mathbf{x}_S, \end{aligned}$$

where

$$\begin{aligned} &\int_{\mathcal{X}_i} \int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_i p_t p_{t'} \cdot (Y_j(i, t) - Y_j(i, t'))^2 dz_t dz_{t'} dz_i \\ &= \int_{\mathcal{X}_i} \int_{\mathcal{X}_{i'}} \int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_i p_{i'} p_t p_{t'} \cdot (Y_j(i, t) - Y_j(i, t'))^2 dz_t dz_{t'} dz_i dz_{i'}. \end{aligned}$$

Let $\alpha(i, t, t')$ denote $Y_j(i, t) - Y_j(i, t')$ and $\beta(i', t, t')$ denote $Y_j(i', t) - Y_j(i', t')$, we can infer that

$$\Delta_t(S', f) - \Delta_t(S, f) = \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \left[\sum_{j \in \mathcal{K}} \frac{1}{4} \left(\int_{\mathcal{X}_i} \int_{\mathcal{X}_{i'}} \int_{\mathcal{X}_i} \int_{\mathcal{X}_{i'}} p_t p_{t'} p_i p_{i'} (\alpha(i, t, t') - \beta(i', t, t'))^2 dz_i dz_{i'} dz_t dz_{t'} \right) \right] d\mathbf{x}_S \geq 0.$$

Consequently, the global sufficient value function v_{suff}^g retains the supermodularity property for any valid classification function, regardless of whether its features are Boolean, discrete, continuous, or a mix thereof.

We now demonstrate that the local probabilistic sufficient value function is neither submodular nor supermodular. As a simple illustration, fix the uniform distribution on the three variables $\{x_1, x_2, x_3\}$ and define a Boolean function as follows:

$$f(x_1, x_2) = (x_1 \vee x_2) \wedge x_3. \quad (52)$$

Consider the input $\mathbf{x} = (1, 1, 1)$ which is classified as 1. For the local probabilistic sufficient value function v_{suff}^l , we have that the following holds:

$$v_{\text{suff}}^l(\emptyset) = \frac{3}{8}, \quad v_{\text{suff}}^l(\{1\}) = \frac{1}{2}, \quad v_{\text{suff}}^l(\{2\}) = \frac{1}{2}, \quad v_{\text{suff}}^l(\{3\}) = \frac{3}{4}, \\ v_{\text{suff}}^l(\{1, 2\}) = \frac{1}{2}, \quad v_{\text{suff}}^l(\{1, 3\}) = 1, \quad v_{\text{suff}}^l(\{2, 3\}) = 1, \quad v_{\text{suff}}^l(\{1, 2, 3\}) = 1.$$

Since $(v_{\text{suff}}^l(\{1, 2\}) - v_{\text{suff}}^l(\{2\})) < (v_{\text{suff}}^l(\{1\}) - v_{\text{suff}}^l(\emptyset)) < (v_{\text{suff}}^l(\{1, 3\}) - v_{\text{suff}}^l(\{3\}))$, the function v_{suff}^l is neither supermodular nor submodular. □

F PROOF OF PROPOSITION 4

Proposition 4. *While the local probabilistic contrastive setting (for any δ) lacks submodularity — even when $\mathcal{D}$ is uniform, i.e., the value function v_{con}^l is not submodular — in the global probabilistic setting (also for any δ), when $\mathcal{D}$ exhibits feature independence, the value function v_{con}^g is submodular.*

Proof. Once we establish that the global sufficient setting is supermodular, it directly follows that the global contrastive setting is submodular. Let $S \subseteq [n]$ be the set of features allowed to change, and suppose $S \subseteq S'$ with $i \notin S'$. To compare the marginal contributions $v_{\text{con}}^g(S \cup \{i\}) - v_{\text{con}}^g(S)$ and $v_{\text{con}}^g(S' \cup \{i\}) - v_{\text{con}}^g(S')$, we equivalently compare $(1 - v_{\text{suff}}^g(\bar{S} \setminus \{i\})) - (1 - v_{\text{suff}}^g(\bar{S}))$ and $(1 - v_{\text{suff}}^g(\bar{S}' \setminus \{i\})) - (1 - v_{\text{suff}}^g(\bar{S}'))$. Since $S \subseteq S'$, it follows that $\bar{S}' \subseteq \bar{S}$, and therefore we obtain:

$$(v_{\text{suff}}^g(\bar{S}) - v_{\text{suff}}^g(\bar{S} \setminus \{i\})) \geq (v_{\text{suff}}^g(\bar{S}') - v_{\text{suff}}^g(\bar{S}' \setminus \{i\})) \quad (53)$$

Which is equivalent to:

$$(v_{\text{con}}^g(S \cup \{i\}) - v_{\text{con}}^g(S)) \geq (v_{\text{con}}^g(S' \cup \{i\}) - v_{\text{con}}^g(S')). \quad (54)$$

Thus, the global contrastive value function v_{con}^g is submodular, hence concluding this proof.

We will now demonstrate that the local probabilistic contrastive value function is neither submodular nor supermodular. As a simple illustration, fix the uniform distribution on the three variables $\{x_1, x_2, x_3\}$ and define a Boolean function as follows:

$$f(x_1, x_2) := (x_1 \vee x_2) \wedge x_3. \quad (55)$$

Consider the input $\mathbf{x} = (1, 1, 1)$ which is classified as 1. For the local probabilistic contrastive value function v_{con}^l , we have:

$$v_{\text{con}}^l(\emptyset) = 1, \quad v_{\text{con}}^l(\{1\}) = 1, \quad v_{\text{con}}^l(\{2\}) = 1, \quad v_{\text{con}}^l(\{3\}) = \frac{1}{2},$$

$$v_{\text{con}}^l(\{1, 2\}) = \frac{3}{4}, \quad v_{\text{con}}^l(\{1, 3\}) = \frac{1}{2}, \quad v_{\text{con}}^l(\{2, 3\}) = \frac{1}{2}, \quad v_{\text{con}}^l(\{1, 2, 3\}) = \frac{3}{8}.$$

Since $(v_{\text{con}}^l(\{1, 2\}) - v_{\text{con}}^l(\{2\})) < (v_{\text{con}}^l(\{1, 2, 3\}) - v_{\text{con}}^l(\{2, 3\})) < (v_{\text{con}}^l(\{1\}) - v_{\text{con}}^l(\emptyset))$, the function v_{con}^l is neither submodular nor supermodular. $\square$

In the rest of this section, we show that the feature independence assumption is not just sufficient but also necessary for supermodularity. We illustrate this with the following counterexample.

Define a Boolean function:

$$f(x_1, x_2) := x_1 \tag{56}$$

For this boolean function, it holds that $f(00) = 0$, $f(01) = 0$, $f(10) = 1$, and $f(11) = 1$. Moreover, define a distribution $\mathcal{D}$ on x_1 and x_2 . Recall that the global sufficient value function is:

$$\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] = \sum_{\mathbf{x} \sim \mathcal{D}} \mathbf{Pr}(\mathbf{x}) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)$$

In addition, if we can show that

$$\Delta_t(S, f) \leq \Delta_t(S', f),$$

where $S \subseteq S'$ and $t \notin S'$, then the global sufficient value function is supermodular. When $S = \emptyset$, we have that the following holds:

$$\begin{aligned} & \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\ &= \mathbf{Pr}(x_1 = 0, x_2 = 0) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0) + \mathbf{Pr}(x_1 = 0, x_2 = 1) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0) + \\ & \quad \mathbf{Pr}(x_1 = 1, x_2 = 0) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1) + \mathbf{Pr}(x_1 = 1, x_2 = 1) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1) \end{aligned}$$

Which is also equivalent to:

$$\begin{aligned} & \mathbf{Pr}(x_1 = 0, x_2 = 0) \cdot (\mathbf{Pr}(x_1 = 0, x_2 = 0) + \mathbf{Pr}(x_1 = 0, x_2 = 1)) + \\ & \quad \mathbf{Pr}(x_1 = 0, x_2 = 1) \cdot (\mathbf{Pr}(x_1 = 0, x_2 = 0) + \mathbf{Pr}(x_1 = 0, x_2 = 1)) + \\ & \quad \mathbf{Pr}(x_1 = 1, x_2 = 0) \cdot (\mathbf{Pr}(x_1 = 1, x_2 = 0) + \mathbf{Pr}(x_1 = 1, x_2 = 1)) + \\ & \quad \mathbf{Pr}(x_1 = 1, x_2 = 1) \cdot (\mathbf{Pr}(x_1 = 1, x_2 = 0) + \mathbf{Pr}(x_1 = 1, x_2 = 1)) \\ &= (\mathbf{Pr}(x_1 = 0, x_2 = 0) + \mathbf{Pr}(x_1 = 0, x_2 = 1))^2 + (\mathbf{Pr}(x_1 = 1, x_2 = 0) + \mathbf{Pr}(x_1 = 1, x_2 = 1))^2 \end{aligned}$$

When $S = \{2\}$, we have that the following condition holds:

$$\begin{aligned} & \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\ &= \mathbf{Pr}(x_1 = 0, x_2 = 0) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0 \mid x_2 = 0) + \\ & \quad \mathbf{Pr}(x_1 = 0, x_2 = 1) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0 \mid x_2 = 1) + \\ & \quad \mathbf{Pr}(x_1 = 1, x_2 = 0) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid x_2 = 0) + \\ & \quad \mathbf{Pr}(x_1 = 1, x_2 = 1) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid x_2 = 1) \end{aligned}$$

This can also be rearranged as follows:

$$\begin{aligned}
& \Pr(x_1 = 0, x_2 = 0) \cdot \frac{\Pr(x_1 = 0, x_2 = 0)}{\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 1, x_2 = 0)} + \\
& \Pr(x_1 = 0, x_2 = 1) \cdot \frac{\Pr(x_1 = 0, x_2 = 1)}{\Pr(x_1 = 0, x_2 = 1) + \Pr(x_1 = 1, x_2 = 1)} + \\
& \Pr(x_1 = 1, x_2 = 0) \cdot \frac{\Pr(x_1 = 1, x_2 = 0)}{\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 1, x_2 = 0)} + \\
& \Pr(x_1 = 1, x_2 = 1) \cdot \frac{\Pr(x_1 = 1, x_2 = 1)}{\Pr(x_1 = 0, x_2 = 1) + \Pr(x_1 = 1, x_2 = 1)} \\
& = \frac{\Pr(x_1 = 0, x_2 = 0)^2 + \Pr(x_1 = 1, x_2 = 0)^2}{\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 1, x_2 = 0)} + \frac{\Pr(x_1 = 0, x_2 = 1)^2 + \Pr(x_1 = 1, x_2 = 1)^2}{\Pr(x_1 = 0, x_2 = 1) + \Pr(x_1 = 1, x_2 = 1)}
\end{aligned}$$

In contrast, when $S = \{1\}$, we have the the following holds:

$$\begin{aligned}
& \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\
& = \Pr(x_1 = 0, x_2 = 0) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0 \mid x_1 = 0) + \\
& \quad \Pr(x_1 = 0, x_2 = 1) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0 \mid x_1 = 0) + \\
& \quad \Pr(x_1 = 1, x_2 = 0) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid x_1 = 1) + \\
& \quad \Pr(x_1 = 1, x_2 = 1) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid x_1 = 1)
\end{aligned}$$

This can similarly be rearranged as follows:

$$\begin{aligned}
& \Pr(x_1 = 0, x_2 = 0) \cdot \frac{\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 0, x_2 = 1)}{\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 0, x_2 = 1)} + \\
& \Pr(x_1 = 0, x_2 = 1) \cdot \frac{\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 0, x_2 = 1)}{\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 0, x_2 = 1)} + \\
& \Pr(x_1 = 1, x_2 = 0) \cdot \frac{\Pr(x_1 = 1, x_2 = 0) + \Pr(x_1 = 1, x_2 = 1)}{\Pr(x_1 = 1, x_2 = 0) + \Pr(x_1 = 1, x_2 = 1)} + \\
& \Pr(x_1 = 1, x_2 = 1) \cdot \frac{\Pr(x_1 = 1, x_2 = 0) + \Pr(x_1 = 1, x_2 = 1)}{\Pr(x_1 = 1, x_2 = 0) + \Pr(x_1 = 1, x_2 = 1)} = 1
\end{aligned}$$

Finally, when $S = \{1, 2\}$, we have the following condition:

$$\begin{aligned}
& \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\
& = \Pr(x_1 = 0, x_2 = 0) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0 \mid x_1 = 0, x_2 = 0) + \\
& \quad \Pr(x_1 = 0, x_2 = 1) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0 \mid x_1 = 0, x_2 = 1) + \\
& \quad \Pr(x_1 = 1, x_2 = 0) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid x_1 = 1, x_2 = 0) + \\
& \quad \Pr(x_1 = 1, x_2 = 1) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid x_1 = 1, x_2 = 1)
\end{aligned}$$

This can likewise be reordered as follows:

$$\begin{aligned}
& \Pr(x_1 = 0, x_2 = 0) \cdot \frac{\Pr(x_1 = 0, x_2 = 0)}{\Pr(x_1 = 0, x_2 = 0)} + \Pr(x_1 = 0, x_2 = 1) \cdot \frac{\Pr(x_1 = 0, x_2 = 1)}{\Pr(x_1 = 0, x_2 = 1)} + \\
& \Pr(x_1 = 1, x_2 = 0) \cdot \frac{\Pr(x_1 = 1, x_2 = 0)}{\Pr(x_1 = 1, x_2 = 0)} + \Pr(x_1 = 1, x_2 = 1) \cdot \frac{\Pr(x_1 = 1, x_2 = 1)}{\Pr(x_1 = 1, x_2 = 1)} \\
& = 1
\end{aligned}$$

Let $t = 2$. Evidently, we have $\Delta_t(\{1\}, f) = 0$. Next, let us analyse $\Delta_t(\emptyset, f)$. We note that if we assume feature independence, we observe that:

$$\begin{aligned}
& (\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 0, x_2 = 1))^2 + (\Pr(x_1 = 1, x_2 = 0) + \Pr(x_1 = 1, x_2 = 1))^2 \\
& = \Pr(x_1 = 0)^2 + \Pr(x_1 = 1)^2,
\end{aligned}$$

Additionally, another condition that holds is:

$$\frac{\Pr(x_1 = 0, x_2 = 0)^2 + \Pr(x_1 = 1, x_2 = 0)^2}{\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 1, x_2 = 0)} + \frac{\Pr(x_1 = 0, x_2 = 1)^2 + \Pr(x_1 = 1, x_2 = 1)^2}{\Pr(x_1 = 0, x_2 = 1) + \Pr(x_1 = 1, x_2 = 1)} = \Pr(x_1 = 0)^2 + \Pr(x_1 = 1)^2.$$

This implies $\Delta_t(\emptyset, f) = 0$, that is, $\Delta_t(\emptyset, f) \leq \Delta_t(\{1\}, f)$. However, without the assumption of feature independence, we cannot guarantee $\Delta_t(\emptyset, f) = 0$. For example, let:

$$\Pr(x_1 = 0, x_2 = 0) = P_{00}, \Pr(x_1 = 0, x_2 = 1) = P_{01}$$

and

$$\Pr(x_1 = 1, x_2 = 0) = P_{10}, \Pr(x_1 = 1, x_2 = 1) = P_{11}$$

Then, by simplifying $\Delta_t(\emptyset, f)$ we get that the following holds:

$$\Delta_t(\emptyset, f) = \left[\frac{P_{00}^2 + P_{10}^2}{P_{00} + P_{10}} + \frac{P_{01}^2 + P_{11}^2}{P_{01} + P_{11}} \right] - \left[(P_{00} + P_{01})^2 + (P_{10} + P_{11})^2 \right] \quad (57)$$

We then let $P_{00} = 0.4$, $P_{01} = 0.2$, $P_{10} = 0.1$, and $P_{11} = 0.3$, and get that the following holds:

$$\Delta_t(\emptyset, f) = \left[\frac{(0.4)^2 + (0.1)^2}{0.5} + \frac{(0.2)^2 + (0.3)^2}{0.5} \right] - \left[(0.4 + 0.2)^2 + (0.1 + 0.3)^2 \right] = 0.08. \quad (58)$$

This implies that $\Delta_t(\emptyset, f) > \Delta_t(1, f)$, indicating that — unlike in the monotonicity setting where it was not required — feature independence is a necessary condition for supermodularity.

G PROOF OF PROPOSITION 5

Proposition 5. *Computing Algorithm 1 with the local value functions v_{con}^ℓ and v_{suff}^ℓ does not always converge to a subset minimal δ -sufficient/contrastive reason. However, computing it with the global value functions v_{con}^g or v_{suff}^g necessarily produces subset minimal δ -sufficient/contrastive reasons.*

Proof. We note that this proposition follows directly from our proof that both v_{suff}^g and v_{con}^g are monotone non-decreasing (as established in Proposition 2), along with other relevant conclusions drawn in prior work. In the remainder of this section, we elaborate on how this monotonicity property leads to the stated corollary.

The convergence of Algorithm 1 to a subset-minimal δ global sufficient or contrastive reason with respect to the value functions v_{con}^g and v_{suff}^g follows directly from the monotonicity property, for which prior work (Ignatiev et al., 2019; Wu et al., 2024b; Arenas et al., 2022) showed that this type of greedy algorithm yields a subset-minimal explanation when the underlying value function is monotone non-decreasing. The algorithm halts when, for all $i \in S$, the value $v(S \setminus \{i\})$ drops below δ , ensuring that although $v(S) \geq \delta$ (a maintained invariant), removing any element causes the condition to fail. Due to monotonicity, this implies that no proper subset of S satisfies the condition, guaranteeing that S is subset-minimal.

We now turn to explain why Algorithm 1 does not converge to a subset-minimal solution in the local setting. We note that it is well known that the local probabilistic sufficient value function (and likewise the local probabilistic contrastive value function) is not monotone Arenas et al. (2022); Izza et al. (2023); Subercaseaux et al. (2025); Izza et al. (2024b). As a simple illustration, fix the uniform distribution on the two binary variables $\{x_1, x_2\}$ and define a Boolean function as follows:

$$f(x_1, x_2) := x_1 \vee x_2. \quad (59)$$

Consider the input $\mathbf{x} = (0, 1)$ which is classified as 1, then:

$$v_{suff}^\ell(\emptyset) = \frac{3}{4}, \quad v_{suff}^\ell(\{1\}) = \frac{1}{2}, \quad v_{suff}^\ell(\{2\}) = 1, \quad v_{suff}^\ell(\{1, 2\}) = 1. \quad (60)$$

Because $v_{\text{suff}}^\ell(\emptyset) > v_{\text{suff}}^\ell(\{1\}) < v_{\text{suff}}^\ell(\{1, 2\})$, the function v_{suff}^ℓ is not monotone. Likewise, it can be computed that:

$$v_{\text{con}}^\ell(\emptyset) = 1, \quad v_{\text{con}}^\ell(\{1\}) = 1, \quad v_{\text{con}}^\ell(\{2\}) = \frac{1}{2}, \quad v_{\text{con}}^\ell(\{1, 2\}) = \frac{3}{4}. \quad (61)$$

(For v_{con}^ℓ , the S denotes the set of features that are allowed to vary.) Because $v_{\text{con}}^\ell(\emptyset) > v_{\text{con}}^\ell(\{2\}) < v_{\text{con}}^\ell(\{1, 2\})$, the function v_{con}^ℓ is also non-monotone.

□

H PROOF OF THEOREM 1

Theorem 1. *If f is a decision tree and the probability term v_{suff}^g can be computed in polynomial time given the distribution $\mathcal{D}$ (which holds for independent distributions, among others), then obtaining a subset-minimal global δ -sufficient reason can be obtained in polynomial time. However, unless $\text{PTIME}=\text{NP}$, no polynomial-time algorithm exists for computing a local δ -sufficient reason for decision trees even under independent distributions.*

Proof. We begin by noting that for decision trees, assuming that $\mathcal{D}$ represents independent distributions, the computation of the value function:

$$v_{\text{suff}}^g(S) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) = f(\mathbf{z}) \mid \mathbf{x}_S = \mathbf{z}_S)] \quad (62)$$

Can be carried out in polynomial time using the following procedure. Thanks to the tractability of decision trees, we can iterate over all pairs of leaf nodes, each corresponding to partial assignments $\mathbf{x}_S$ and $\mathbf{z}_{S'}$. For each such pair, if both leaf nodes yield the same prediction under f , we compute the corresponding term in the expected value $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) = f(\mathbf{z}) \mid \mathbf{x}_S = \mathbf{z}_S)]$ by multiplying the respective feature-wise probabilities over the shared features of the two vectors. Under the feature independence assumption, these probabilities decompose, allowing the full expectation to be computed by summing the contributions of all such matching leaf pairs.

Since all probabilities involved are provided as part of the input and each step involves only polynomial-time operations, the entire procedure runs in polynomial time. This establishes that every step in Algorithm 1 is efficient, and thus the algorithm as a whole runs in polynomial time. Finally, combining this with Lemma 2, which proves that v_{suff}^g is monotone non-decreasing, we conclude that Algorithm 1 converges to a subset-minimal explanation. This is because the algorithm halts when, for all $i \in S$, the value $v(S \setminus \{i\})$ falls below δ , while $v(S) \geq \delta$ is preserved throughout. By monotonicity, this guarantees that no strict subset of S satisfies the condition, ensuring S is indeed subset-minimal.

For the remaining part of the claim, the result follows from (Arenas et al., 2022), which showed that, assuming $\text{P} \neq \text{NP}$, there is no polynomial-time algorithm for computing a subset-minimal δ -sufficient reason for decision trees. While this was proven specifically under the uniform distribution, the hardness clearly extends to independent distributions, which include this case. Combined, these results establish the complexity separation between the local and global variants, thus concluding our proof.

□

I PROOF OF PROPOSITION 6

Proposition 6. *For any model f , and empirical distribution $\mathcal{D}$ — computing a subset-minimal global δ -sufficient or δ -contrastive reason for f can be done in polynomial time.*

Proof. The computation of the global sufficient probability function:

$$v_{\text{suff}}^g(S) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) = f(\mathbf{z}) \mid \mathbf{x}_S = \mathbf{z}_S)] \quad (63)$$

or the computation of the global contrastive probability function

$$v_{\text{con}}^g(S) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) \neq f(\mathbf{z}) \mid \mathbf{x}_{\bar{S}} = \mathbf{z}_{\bar{S}})] \quad (64)$$

Can be performed in polynomial time when $\mathcal{D}$ is selected from the class of empirical distributions. This is achieved by iterating over pairs of instances $\mathbf{x}, \mathbf{z}$ within the dataset and running an inference through $f(\mathbf{x})$ and $f(\mathbf{z})$ to compute the expected values at each step. Consequently, determining whether the probability function exceeds or falls below a given threshold δ can also be accomplished in polynomial time. Furthermore, leveraging the proof of non-decreasing monotonicity (Proposition D) of the global probability function, we can utilize algorithm 1 for contrastive reason search or for sufficient reason search. By executing a linear number of polynomial queries, a subset-minimal explanation (whether a subset minimal contrastive reason or a subset minimal sufficient reason) can thus be obtained in polynomial time.

□

J PROOF OF THEOREM 2

Theorem 2. *Assuming f is a neural network or a tree ensemble, and $\mathcal{D}$ is an empirical distribution — there exist polynomial-time algorithms for obtaining subset minimal global δ -sufficient and contrastive reasons. However, unless $\text{PTIME}=\text{NP}$, there is no polynomial time algorithm for computing a subset minimal local δ -sufficient reason or a subset minimal local δ -contrastive reason.*

Proof. The first part of the proof follows directly from Proposition 6, which established that this holds for any model, since both $v_{\text{suff}}^g(S) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) = f(\mathbf{z}) \mid \mathbf{x}_S = \mathbf{z}_S)]$ and $v_{\text{con}}^g(S) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) \neq f(\mathbf{z}) \mid \mathbf{x}_{\bar{S}} = \mathbf{z}_{\bar{S}})]$ can be computed in polynomial time from the empirical distribution. For the second part, we proceed by proving two lemmas: one establishing the intractability of computing the sufficient case, and the other addressing the contrastive case.

Lemma 6. *Unless $\text{PTIME}=\text{NP}$, then there is no polynomial time algorithm for computing a subset minimal local δ -sufficient reason for either a tree ensemble or a neural network, under empirical distributions.*

Proof. We will first prove this claim for neural networks and then extend the result to tree ensembles. We will establish this claim by demonstrating that if a polynomial-time algorithm exists for computing a subset-minimal local δ sufficient reason for neural networks, then it would enable us to solve the classic NP-complete CNF-SAT problem in polynomial time. The CNF-SAT problem is defined as follows:

CNF-SAT:

Input: A formula in conjunctive normal form (CNF): ϕ .

Output: Yes, if there exists an assignment to the n literals of ϕ such that ϕ is evaluated to True, and No otherwise

Our proof will also utilize the following lemma (with its proof provided in Barceló et al. (2020)):

Lemma 7. *Any boolean circuit ϕ can be encoded into an equivalent MLP over the binary domain $\{0, 1\}^n \rightarrow \{0, 1\}$ in polynomial time.*

Proof. We will actually establish hardness for a simpler, more specific case, which will consequently imply hardness for the more general setting. In this case, we assume that the empirical distribution $\mathcal{D}$ consists of a single element, which, for simplicity, we take to be the zero vector $\mathbf{0}_n$. Since the sufficiency of S in this scenario depends only on the local instance $\mathbf{x}$ and the single instance $\mathbf{0}_n$ in our empirical distribution, the nature of the input format (whether discrete, continuous, etc.) does not affect the result. Therefore, the hardness results hold universally across all these settings.

Similarly to $\mathbf{0}_n$, we define $\mathbf{1}_n$ as an n -dimensional vector consisting entirely of ones. Given an input ϕ , we initially assign $\mathbf{1}_n$ to ϕ , effectively setting all variables to True. If ϕ evaluates to True under this assignment, then a satisfying truth assignment exists. Therefore, we assume that $\phi(\mathbf{1}_n) = 0$. We now introduce the following definitions:

$$\begin{aligned}\phi_2 &:= (x_1 \wedge x_2 \wedge \dots \wedge x_n), \\ \phi' &:= \phi \vee \phi_2\end{aligned}\tag{65}$$

ϕ' , while no longer a CNF, can still be transformed into an MLP f using Lemma 8, ensuring that f behaves equivalently to ϕ' over the domain $\{0, 1\}^n$. Given our assumption that computing a subset-minimal local δ -sufficient reason is feasible in polynomial time, we can determine one for the instance $\langle f, \mathbf{x} := \mathbf{1}_n \rangle$, $\delta := 1$, noting that the empirical distribution $\mathcal{D}$ is simply defined over the single data point $\mathbf{0}_n$.

We now assert that the subset-minimal δ -sufficient reason generated for $\langle f, \mathbf{x} \rangle$ encompasses the entire input space, i.e., $S = \{1, \dots, n\}$, if and only if $\langle \phi \rangle \notin \text{CNF-SAT}$.

Let us assume that $S = \{1, \dots, n\}$. Since S is a δ -sufficient reason for $\langle f, \mathbf{x} \rangle$, this simply means that setting the complementary set to any value maintains the prediction. Since the complementary set is $\emptyset$ in this case, this trivially holds. The fact that S is subset-minimal means that any other subset $S' \subseteq S$ satisfies $v(S') < \delta = 1$. Since the probability function $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) = f(\mathbf{z}) \mid \mathbf{x}_S = \mathbf{z}_S)$ is determined by a single point (the distribution contains only the point $\mathbf{0}_n$), the probability function can only take values of 1 or 0. Hence, we also know that $v(S') = 0$. This tells us that, aside from the subset $S = \{1, \dots, n\}$, for any subset $S' \subseteq S$, fixing the features in S' to 1 and the rest to 0 does not result in a classification outcome of 1. Since the ϕ_2 component within ϕ' is True only if all features are assigned 1, this directly implies that ϕ is assigned False for any of these inputs. Since we already know that ϕ does not return a True answer for the vector assignment $\mathbf{1}_n$ (as verified at the beginning), and now we have established that the same holds for all other input vectors, we conclude that $\phi \notin \text{CNF-SAT}$.

Now, suppose that S is a subset that is strictly contained within $\{1, \dots, n\}$. Given that S is sufficient under our definitions of the distribution $\mathcal{D}$ with $\delta = 1$, we can apply the same reasoning as before to conclude that $v(S) = 1$. This implies that setting the features in S to 1 while setting the remaining features to 0 ensures that the function f evaluates to 1. Since this assignment is necessarily not a vector consisting entirely of ones, it follows that the ϕ_2 component within ϕ' must be False. Consequently, the ϕ component must be True, which implies that $\langle \phi \rangle \in \text{CNF-SAT}$. This completes the proof.

We have established that the following claim holds for neural networks. However, extending the proof to tree ensembles requires a minor and straightforward adaptation. To do so, we will utilize the following lemma, which has been noted in several previous works (Ordyniak et al., 2024; Audemard et al., 2022b):

Lemma 8. *Any CNF or DNF ϕ can be encoded into an equivalent random forest classifier over the binary domain $\{0, 1\}^n \rightarrow \{0, 1\}$ in polynomial time.*

Proof. We observe that we can apply the same process used in our proof for neural networks, where we encoded ϕ' into an equivalent neural network. However, ϕ' is no longer a valid CNF due to our construction (though encoding it into an MLP was not an issue, as any Boolean circuit can be transformed into an MLP). Nevertheless, since ϕ' consists of a conjunction of only two terms, we can easily represent it as an equivalent CNF:

$$\phi' := (c_1 \vee \phi_2) \wedge (c_2 \vee \phi_2) \wedge \dots \wedge (c_m \vee \phi_2)\tag{66}$$

Where each c_i is a disjunction of a few terms. Consequently, ϕ' is a valid CNF, allowing us to transform it into an equivalent random forest classifier. The reduction we outlined for MLPs applies directly to these models as well, thereby completing the proof for both model families.

□

Lemma 9. *Unless $\text{PTIME} = \text{NP}$, then there is no polynomial time algorithm for computing a subset minimal local δ -contrastive reason for either a tree ensemble or a neural network, under empirical distributions.*

Proof. We will present a proof analogous to the one in Lemma 6. Specifically, we will once again utilize the classical NP-hard CNF-SAT problem defined in Lemma 6. In particular, given a Boolean formula ϕ , we will demonstrate that determining a subset-minimal contrastive reason — whether for a neural network or a tree ensemble — allows us to decide the satisfiability of ϕ .

First, we check whether assigning all variables in ϕ to 1 evaluates ϕ to True. If so, the formula is satisfiable, and we have determined its satisfiability. Otherwise, we use Lemma 8 to encode the CNF formula as a neural network f . Next, we compute a subset-minimal δ -sufficient reason by setting $\mathcal{D}$ as the empirical distribution containing only a single data point $\mathbf{0}_n$, following a similar procedure to Lemma 6. Additionally, we set $\delta = 1$ and compute a subset-minimal δ -contrastive reason concerning $\langle f, \mathbf{x} \rangle$.

We will now demonstrate that if any subset-minimal δ contrastive reason obtained for ϕ is satisfiable, then ϕ itself is satisfiable. Conversely, if no subset-minimal δ contrastive reason is obtained, then ϕ is unsatisfiable. The validity of this claim follows a reasoning similar to that provided in Lemma 6. Specifically, the term $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) \neq f(\mathbf{z}) \mid \mathbf{x}_{\bar{S}} = \mathbf{z}_{\bar{S}})$, where the distribution $\mathcal{D}$ considers sampling from a single datapoint, can set the probability to either 0 or 1. Furthermore, since we are searching for a $\delta = 1$ contrastive reason, this is equivalent to asking whether there exists an assignment that changes the classification of f , which corresponds to modifying the assignment of ϕ from False to True. If such an assignment exists, then ϕ is satisfiable. However, if no subset-minimal contrastive reason exists, then no subset of features fixed to zero — when the complementary set is set to ones — evaluates to true. This is equivalent to stating that no assignment evaluates f (and consequently ϕ) to True, implying that ϕ is unsatisfiable.

To extend the proof from neural networks to tree ensembles, we can follow the same procedure outlined in Lemma 6, encoding the CNF formula into an equivalent random forest classifier. Consequently, the proof remains valid for tree ensembles, thereby concluding the proof.

□

K PROOF OF THEOREM 3

Theorem 3. *Given a neural network or tree ensemble f and an empirical distribution $\mathcal{D}$ over a fixed dataset D , Algorithm 2 yields a constant $\mathcal{O}\left(\ln\left(\frac{v_{\text{con}}^g([n])}{\min_{i \in [n]} v_{\text{con}}^g(\{i\})}\right)\right)$ -approximation, bounded by $\mathcal{O}(\ln(|D|))$, for computing a global cardinally minimal δ -contrastive reason for f , assuming feature independence. In contrast, unless $\text{PTIME}=\text{NP}$, no bounded approximation exists for computing a local cardinally minimal δ -contrastive reason for any $\langle f, \mathbf{x} \rangle$, even when $|D| = 1$.*

Proof. We divide the proof of the theorem into two lemmas, covering both the approximation guarantee and the result on the absence of a bounded approximation.

Lemma 10. *Given a neural network or tree ensemble f and an empirical distribution $\mathcal{D}$ over a fixed dataset D , Algorithm 2 yields a constant $\mathcal{O}\left(\ln\left(\frac{v_{\text{con}}^g([n])}{\min_{i \in [n]} v_{\text{con}}^g(\{i\})}\right)\right)$ -approximation, bounded by $\mathcal{O}(\ln(|D|))$, for computing a global cardinally minimal δ -contrastive reason for f , assuming feature independence.*

Proof. We will in fact prove a stronger claim, showing that this holds for *any* model, provided the trivial condition that its inference time is computable in polynomial time, along with one additional mild condition that we will detail later — both of which apply to both our neural network and tree ensemble formalizations.

We begin by noting that, since we are working with empirical distributions, the computation of the global contrastive probability value function:

$$v_{\text{con}}^g(S) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) \neq f(\mathbf{z}) \mid \mathbf{x}_{\bar{S}} = \mathbf{z}_{\bar{S}})] \quad (67)$$

Can be computed in polynomial time by iterating over all pairs $\mathbf{x}, \mathbf{z}$ in the dataset D , as previously established in Proposition 2. Since Algorithm 2 performs only a linear number of these polynomial-time queries, its total runtime is therefore polynomial.

Regarding the approximation, the classical work by Wolsey et al. (Wolsey, 1982) established a harmonic-series-based approximation guarantee for monotone non-decreasing submodular functions v with integer values. More generally, their result yields an approximation factor of $\mathcal{O}\left(\ln\left(\frac{v([n])}{\min_{i \in [n]} v(\{i\})}\right)\right)$. We showed in Proposition 2 that the value function v_{con}^g is monotone

non-decreasing, and under feature independence, Proposition 4 establishes its submodularity. Combined with the fact that Algorithm 2 runs in polynomial time, this directly yields an approximation guarantee of $\mathcal{O}\left(\ln\left(\frac{v_{\text{con}}^g([n])}{\min_{i \in [n]} v_{\text{con}}^g(\{i\})}\right)\right)$.

However, we must also show that the expression $\mathcal{O}\left(\ln\left(\frac{v_{\text{con}}^g([n])}{\min_{i \in [n]} v_{\text{con}}^g(\{i\})}\right)\right)$ is both finite and bounded by $\mathcal{O}(\ln(|D|))$, implying that it is effectively constant, due to the assumption of a fixed dataset $|D|$. To ensure this, we begin by confirming that the expression is well-defined and finite. We then proceed to establish the desired bounds. To do so, we introduce a preprocessing step in which we eliminate “redundant” elements—those that could theoretically cause the denominator $\min_{i \in [n]} v_{\text{con}}^g(\{i\})$ to be zero. We begin by formally defining what we mean by redundancy:

Definition 1. Let $\mathcal{D}$ denote some empirical distribution over a dataset D . Then we say that some feature $i \in [n]$ is redundant with respect to $\mathcal{D}$ if for any pair $\mathbf{x}, \mathbf{z} \in D$ it holds that $f(\mathbf{x}_{[n] \setminus \{i\}}; \mathbf{z}_{\{i\}}) = f(\mathbf{x})$.

Here, the notation $f(\mathbf{x}_{[n] \setminus \{i\}}; \mathbf{z}_{\{i\}})$ indicates that all features in $[n] \setminus \{i\}$ are fixed to their values in $\mathbf{x}$, while feature i is set to its value in $\mathbf{z}$. Notably, this is equivalent to defining:

$$\forall S \subseteq [n], \mathbf{z} \in D \quad f(\mathbf{x}_S; \mathbf{z}_{\bar{S}}) = f(\mathbf{x}_{S \setminus \{i\}}; \mathbf{z}_{\bar{S} \cup \{i\}}) \quad (68)$$

As before, the notation $(\mathbf{x}_S; \mathbf{z}_{\bar{S}})$ indicates that the features in S are fixed to their values in $\mathbf{x}$, while the features in $\bar{S}$ are fixed to their values in $\mathbf{z}$. Thus, in this sense, if we “remove” feature i from the input space and define a new function f' over the reduced space $[n'] := [n] \setminus \{i\}$, then for any input of size n' , the output of f' will exactly match the output of f when applied to the same features $[n] \setminus \{i\}$. This removal can be done in polynomial time for both tree ensembles and neural networks: for neural networks, it involves detaching the corresponding input neuron from the network; for tree ensembles, it involves removing any splits on that feature from all decision trees. Given the empirical nature of the contrastive value function $v_{\text{con}}^g(S)$ — as previously discussed in this proof and in Proposition 2 — we can compute each $v_{\text{con}}^g(\{i\})$ for all $i \in [n]$ by iterating over all pairs $\mathbf{x}, \mathbf{z} \in D$ and checking whether:

$$f(\mathbf{x}_{[n] \setminus \{i\}}; \mathbf{z}_{\{i\}}) \neq f(\mathbf{x}) \quad (69)$$

If this condition holds, it indicates that feature i is not redundant. Conversely, if the condition fails for all pairs considered during iteration, then i is deemed redundant. Once all redundant features have been identified with respect to the empirical distribution $\mathcal{D}$, we can remove them and construct an equivalent model f' . As discussed above, this transformation can be performed in polynomial time for both neural networks and decision trees.

To conclude our proof, we observe that $v_{\text{con}}^g(\{i\})$ equals 0 for some empirical distribution $\mathcal{D}$ if and only if feature i is redundant with respect to $\mathcal{D}$. Therefore, once the preprocessing step removes all redundant features from f and we construct the resulting function f' , we ensure that $\min_{i \in [n]} v_{\text{con}}^g(\{i\}) > 0$.

We now present the more precise bounds referenced in our proof. Specifically, we demonstrate that the approximation factor is $\mathcal{O}(\ln(|D|))$. Given that the empirical distribution D is fixed, this yields a constant approximation. While tighter bounds may be achievable, our goal here is solely to establish that the bound is *constant* — a property that will later sharply contrast with the local explainability setting, where no bounded approximation exists.

More specifically, we know that the probability function $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) \neq f(\mathbf{z}) \mid \mathbf{x}_{\bar{S}} = \mathbf{z}_{\bar{S}})$ is, by the definition of empirical distributions, at least $\frac{1}{|D|}$. Therefore, the definition of $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) \neq f(\mathbf{z}) \mid \mathbf{x}_{\bar{S}} = \mathbf{z}_{\bar{S}})]$ (i.e., the value function v_{con}^g) is at least $\frac{1}{|D|^2}$. In particular, this also implies that:

$$\min_{i \in [n]} v_{\text{con}}^g(\{i\}) \geq \frac{1}{|D|^2} \quad (70)$$

Consequently, we obtain that:

$$\ln \left(\frac{v_{con}^g([n])}{\min_{i \in [n]} v_{con}^g(\{i\})} \right) \leq \ln(|D|^2) = \mathcal{O}(\ln(|D|)) \quad (71)$$

Which concludes our proof.

□

Lemma 11. *Unless $PTIME=NP$, there is no bounded approximation for computing a cardinally minimal local δ -contrastive reason for any $\langle f, \mathbf{x} \rangle$ where f is either a neural network or a tree ensemble, even when $|D| = 1$.*

Proof. Assume, for contradiction, that there exists a bounded approximation algorithm for computing a cardinally minimal local δ -contrastive reason for some $\langle f, \mathbf{x} \rangle$, where f is a neural network or a tree ensemble—even when $|D| = 1$. However, Lemma 9 establishes that even with a single baseline $\mathbf{z} = 0_n$ (i.e., when the entire dataset is just one instance), deciding whether a contrastive reason exists is NP-hard unless $PTIME = NP$. Therefore, if a polynomial-time algorithm could yield a bounded approximation for a cardinally minimal contrastive reason, it would contradict this result, as such an approximation would implicitly decide the existence of a contrastive explanation.

□

L PROOF OF THEOREM 4

Theorem 4. *Given a neural network or tree ensemble f and an empirical distribution $\mathcal{D}$ over a fixed dataset D , Algorithm 2 yields a constant $\mathcal{O} \left(\frac{1}{1-k^f} + \ln \left(\frac{v_{suff}^g([n])}{\min_{i \in [n]} v_{suff}^g(\{i\})} \right) \right)$ -approximation for computing a global cardinally minimal δ -sufficient reason for f , assuming feature independence. In contrast, unless $PTIME=NP$, there is no bounded approximation for computing a local cardinally minimal δ -sufficient reason for any $\langle f, \mathbf{x} \rangle$, even when $|D| = 1$.*

Proof. We divide the proof into two lemmas: one establishing the approximation guarantee for the global case, and the other demonstrating the intractability of the local case.

Lemma 12. *Given a neural network or tree ensemble f and an empirical distribution $\mathcal{D}$ over a fixed dataset D , Algorithm 2 yields a constant $\mathcal{O} \left(\frac{1}{1-k^f} + \ln \left(\frac{v_{suff}^g([n])}{\min_{i \in [n]} v_{suff}^g(\{i\})} \right) \right)$ -approximation for computing a global cardinally minimal δ -sufficient reason for f , assuming feature independence.*

Proof. The proof will follow a similar approach to that of Lemma 10, where we showed that for both neural networks and tree ensembles, Algorithm 2 achieves an approximation factor of $\ln \left(\frac{v_{con}^g([n])}{\min_{i \in [n]} v_{con}^g(\{i\})} \right)$. After applying the preprocessing step, this ratio is guaranteed to be finite and bounded by $\mathcal{O}(\ln(|D|))$.

Similar to Lemma 10, we will again prove a stronger claim — namely, that the result holds for *any* model, assuming the trivial condition that its inference time is polynomially computable, and the additional condition concerning the removal of redundant features, as described in Lemma 10. As discussed there, both conditions are satisfied by our neural network and tree ensemble formalizations.

Here as well, since we are working with empirical distributions, the computation of the global sufficient probability value function:

$$v_{suff}^g(S) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}} [\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}} (f(\mathbf{x}) = f(\mathbf{z}) \mid \mathbf{x}_S = \mathbf{z}_S)] \quad (72)$$

can be computed in polynomial time by iterating over all pairs $\mathbf{x}, \mathbf{z}$ in the dataset D , as previously shown in both Lemma 10 and Proposition 2. Since Algorithm 2 makes only a linear number of such polynomial-time queries, its overall runtime is polynomial.

Unlike Lemma 10, where the approximation relied on the result by Wolsey et al. (Wolsey, 1982) for monotone non-decreasing submodular functions, the setting here requires a different condition due to the supermodular nature of the function. Specifically, Shi et al. (Shi et al., 2021) provided an approximation guarantee of $\mathcal{O} \left(\frac{1}{1-k^f} + \ln \left(\frac{v_{suff}^g([n])}{\min_{i \in [n]} v_{suff}^g(\{i\})} \right) \right)$, where $k^f := 1 - \min_{i \in [n]} \frac{v([n]) - v([n] \setminus i)}{v(i) - v(\emptyset)}$.

We established in Proposition 2 that the value function v_{suff}^g is monotone non-decreasing, and under the assumption of feature independence, Proposition 3 further shows that it is supermodular. Given that Algorithm 2 runs in polynomial time, this leads directly to an approximation guarantee of $\mathcal{O}\left(\frac{1}{1-k^f} + \ln\left(\frac{v_{\text{suff}}^g([n])}{\min_{i \in [n]} v_{\text{suff}}^g(\{i\})}\right)\right)$, where $k^f := 1 - \min_{i \in [n]} \frac{v_{\text{suff}}^g([n]) - v_{\text{suff}}^g([n] \setminus \{i\})}{v_{\text{suff}}^g(\{i\}) - v_{\text{suff}}^g(\emptyset)}$.

To show that this expression is both bounded and constant, we follow the same preprocessing step as in Lemma 10, where we remove all redundant features from f — a process that, as previously explained, can be carried out in polynomial time for both neural networks and tree ensembles. As before, a feature i is strictly redundant if and only if $v(\{i\}) - v(\emptyset) = 0$. This preprocessing yields a new function f' that behaves identically to f over the remaining features and ensures that, for every i , both $v_{\text{suff}}^g(\{i\}) > 0$ and $v_{\text{suff}}^g(\{i\}) - v_{\text{suff}}^g(\emptyset) > 0$ hold.

In order for us to show that this expression is both bounded and constant, similarly to Lemma 10 we will perform the exact same preprocessing phase (which, as we explained there, can be performed in polynomial time both for neural networks as well as tree ensembles) as before where we remove all redundant features from f . We note that here too, it holds that a feature i is strictly redundant if and only if $v(\{i\}) - v(\emptyset) = 0$. Hence, this preprocessing phase will give us a new function f' for which it both holds that it is equivalent to the behaviour of f for all remaining features and also satisfies that for any i it holds that both $v_{\text{suff}}^g(\{i\}) > 0$ and $v_{\text{suff}}^g(\{i\}) - v_{\text{suff}}^g(\emptyset) > 0$ hold.

Now, following the same reasoning as in Lemma 10, where the probability term is lower bounded by $\frac{1}{|D|}$ and its expected value by $\frac{1}{|D|^2}$, we obtain that $\min_{i \in [n]} v_{\text{suff}}^g(i) \geq \frac{1}{|D|^2}$ and $v_{\text{suff}}^g(i) - v_{\text{suff}}^g(\emptyset) \geq \frac{1}{|D|^2}$. This implies, as in Lemma 10, that the first term is lower bounded by $\mathcal{O}(\ln(|D|))$, and for the second term we derive the following:

Since v_{suff}^g is supermodular (Proposition 3), we have that $v_{\text{suff}}^g([n]) - v_{\text{suff}}^g([n] \setminus \{i\})$. It follows that $1 \leq v_{\text{suff}}^g([n]) - v_{\text{suff}}^g([n] \setminus \{i\}) \leq \frac{1}{|D|^2}$. This implies: $1 \leq \min_{i \in [n]} \frac{v_{\text{suff}}^g([n]) - v_{\text{suff}}^g([n] \setminus \{i\})}{v_{\text{suff}}^g(\{i\}) - v_{\text{suff}}^g(\emptyset)} \leq |D|^2$, and therefore, $1 - |D|^2 \leq k^f \leq 0$. This yields $\frac{1}{1-k^f} \leq |D|^2$, demonstrating that the overall approximation bound based on the one established by Shi et al. is constant. While this bound may be significantly smaller in practice, our goal here is simply to show that it remains constant — unlike in the local setting, where no bounded approximation is achievable.

□

Lemma 13. *Unless $\text{PTIME} = \text{NP}$, there is no bounded approximation for computing a local cardinally minimal δ -sufficient reason for any $\langle f, \mathbf{x} \rangle$, even when $|D| = 1$.*

Proof. The proof follows a similar approach to Lemma 11. Suppose, for contradiction, that there exists a polynomial-time algorithm that provides a bounded approximation to a cardinally minimal local δ -sufficient reason for some $\langle f, \mathbf{x} \rangle$, where f is either a neural network or a tree ensemble—even when $|D| = 1$. Yet, Lemma 6 shows that even in the extreme case where the dataset consists of a single baseline $\mathbf{z} = 0_n$, deciding whether a sufficient reason exists is NP-hard unless $\text{PTIME} = \text{NP}$. Hence, the existence of such an approximation algorithm would contradict this hardness result, as it would entail the ability to decide the existence of a sufficient reason.

□

M HARDNESS OF APPROXIMATING GLOBAL CONTRASTIVE REASONS FOR NEURAL NETWORKS USING EMPIRICAL DISTRIBUTIONS

Proposition 1. *Assume a neural network $f : \mathbb{F} \rightarrow [c]$ with ReLU activations, some dataset $\mathbf{D}$, such that $\mathcal{D}$ is the empirical distribution defined over $\mathbf{D}$, and some threshold $\delta \in \mathbb{Q}$. Then unless $\text{NP} \subseteq \text{DTIME}(n^{\mathcal{O}(\log(\log(n)))})$, there does not exist a polynomial-time $\mathcal{O}(\ln(|\mathbf{D}|) - \epsilon)$ approximation algorithm for computing a cardinally minimal global contrastive reason for f concerning $\mathcal{D}$, for any fixed $\epsilon > 0$.*

Proof. The proof proceeds via a polynomial-time approximation-preserving reduction from a specific variant of Set Cover that is, similarly to regular Set Cover, provably hard to approximate within the factor given in the proposition (Vazirani, 2001).

Partial Set Cover:

Input: A universe $U := \{u_1, \dots, u_n\}$, a collection of sets $S := \{S_1, \dots, S_m\}$ such that for all j $S_j \subseteq U$, and a coverage threshold $\delta \in \{0, 1, \dots, n\}$.

Output: A minimum-size subset $C \subseteq S$ such that $|\cup_{S_j \in C} S_j| \geq \delta$.

We reduce this problem to that of computing a *global*, cardinality-minimal contrastive reason S for a neural network f with respect to a distribution $\mathcal{D}$ defined over a dataset $\mathbf{D}$, and some threshold δ' . We begin by recalling the definition of a global contrastive reason and then extend it to the setting where the explanation is evaluated over an empirical dataset $\mathbf{D}$. The definition is as follows:

$$\mathbb{E}_{\mathbf{x} \sim \mathcal{D}} [\Pr_{\mathbf{z} \sim \mathcal{D}} (f(\mathbf{z}) \neq f(\mathbf{x}) \mid \mathbf{z}_{\bar{S}} = \mathbf{x}_{\bar{S}})] \geq \delta. \quad (73)$$

More specifically, when we assume an empirical distribution over $\mathbf{D}$, this implies that:

$$\begin{aligned} \mathbb{E}_{\mathbf{x} \sim \mathcal{D}} [\Pr_{\mathbf{z} \sim \mathcal{D}} (f(\mathbf{z}) \neq f(\mathbf{x}) \mid \mathbf{z}_{\bar{S}} = \mathbf{x}_{\bar{S}})] &= \\ \frac{1}{|\mathbf{D}|} \sum_{\mathbf{x} \in \mathbf{D}} [\Pr_{\mathbf{z} \sim \mathcal{D}} (f(\mathbf{z}) \neq f(\mathbf{x}) \mid \mathbf{z}_{\bar{S}} = \mathbf{x}_{\bar{S}})] &= \\ \frac{1}{|\mathbf{D}|} \sum_{\mathbf{x} \in \mathbf{D}} \left(\frac{\sum_{\mathbf{z} \in \mathbf{D}_{\bar{S}}(\mathbf{z}, \mathbf{x})} \mathbf{1}_{\{f(\mathbf{z}) \neq f(\mathbf{x})\}}}{\mathbf{D}_{\bar{S}}(\mathbf{z}, \mathbf{x})} \right) \end{aligned} \quad (74)$$

where we denote $\mathbf{D}_{\bar{S}}(\mathbf{z}, \mathbf{x}) := \{\mathbf{z} \in \mathbf{D} : \mathbf{z}_{\bar{S}} = \mathbf{x}_{\bar{S}}\}$. We now construct a ReLU neural network f . Before doing so, we specify the functional form that we intend the constructed neural network f to implement. For each set S_j , we introduce two vectors $\mathbf{x}^j$ and $\mathbf{y}^j$, both belonging to $\mathbb{Z}^n$ and drawn from the dataset $\mathbf{D}$, such that they agree on all coordinates outside S_j ; that is, they share identical feature values on the complement $\bar{S}_j$. Formally:

$$\mathbf{x}_{\bar{S}_j}^j = \mathbf{y}_{\bar{S}_j}^j \quad (75)$$

and we assign strictly different values to the features in S_j by selecting them from two distinct vectors $\mathbf{z}$ and $\mathbf{z}'$ such that:

$$\mathbf{x}^j = (\mathbf{x}_{\bar{S}_j}^j; \mathbf{z}_{S_j}^j) \neq (\mathbf{x}_{\bar{S}_j}^j; \mathbf{z}'_{S_j}) = (\mathbf{y}_{\bar{S}_j}^j; \mathbf{z}'_{S_j}) = \mathbf{y}^j \quad (76)$$

We aim to construct f such that it satisfies the following:

$$f(\mathbf{x}^j) = f(\mathbf{x}_{\bar{S}_j}^j; \mathbf{z}_{S_j}^j) \neq f(\mathbf{x}_{\bar{S}_j}^j; \mathbf{z}'_{S_j}) = f(\mathbf{y}_{\bar{S}_j}^j; \mathbf{z}'_{S_j}) = f(\mathbf{y}^j) \quad (77)$$

Let us now assume that both $\mathbf{x}^j$ and $\mathbf{y}^j$ occur *uniquely* in the dataset $\mathbf{D}$ across all $[n]$ coordinates — that is, no other vector in $\mathbf{D}$ matches either of them on any coordinate. Under this assumption, we obtain the following relation:

$$\begin{aligned} \mathbb{E}_{\mathbf{x} \sim \mathcal{D}} [\Pr_{\mathbf{z} \sim \mathcal{D}} (f(\mathbf{z}) \neq f(\mathbf{x}) \mid \mathbf{z}_{\bar{S}_j} = \mathbf{x}_{\bar{S}_j})] &= \\ \frac{1}{|\mathbf{D}|} \sum_{\mathbf{x} \in \mathbf{D}} \left(\frac{\sum_{\mathbf{z} \in \mathbf{D}_{\bar{S}_j}(\mathbf{z}, \mathbf{x})} \mathbf{1}_{\{f(\mathbf{z}) \neq f(\mathbf{x})\}}}{\mathbf{D}_{\bar{S}_j}(\mathbf{z}, \mathbf{x})} \right) &= \frac{2}{2|\mathbf{D}|} = \frac{1}{|\mathbf{D}|} \end{aligned} \quad (78)$$

Here, the factor of 2 appearing in both the numerator and denominator arises from the fact that only two vector pairs — $(\mathbf{x}^j, \mathbf{z}^j)$ and $(\mathbf{z}^j, \mathbf{x}^j)$ — contribute to the respective sums.

We repeat this construction for each S_i in the set-cover instance, each time choosing two *distinct* vectors $\mathbf{x}^i$ and $\mathbf{y}^i$ that differ on *every* coordinate in $[n]$. In total, this yields $2m$ vectors that form the dataset $\mathbf{D}$ from which we construct f .

We then construct a neural network f that simulates this function. This is achieved using a known result on *neural network memorization* (Vardi et al., 2022), which guarantees that a ReLU network

capable of representing any finite set of input–output pairs can be built in polynomial time. Using this result, we construct f accordingly, and set $\delta' := \frac{\delta}{|\mathcal{D}|}$.

We now argue that a subset S_j is a global contrastive reason for f under tolerance δ' if and only if C is a partial set cover for U under tolerance δ . By Equation 78, each constructed pair of vectors $(\mathbf{x}^j, \mathbf{y}^j)$ contributes exactly $\frac{1}{|\mathcal{D}|}$ to the value term. Moreover, because each S_j uses its own distinct pair of vectors in $\mathbb{Z}^n$, there is no overlap between these pairs. Consequently, the coverage achieved by any set cover C corresponds exactly to the accumulated contribution of the selected subsets in the input domain of f . This completes the reduction.

□

N TRACTABILITY RESULTS FOR ORTHOGONAL DNFs

Lemma 14. *If f is an orthogonal DNFs and the probability term v_{suff}^g can be computed in polynomial time given the distribution $\mathcal{D}$ (which holds for independent distributions, among others), then obtaining a subset-minimal global δ -sufficient reason can be obtained in polynomial time.*

Proof. We first state the orthogonal-DNF representation (Crama & Hammer, 2011) used by the proof.

Definition 2 (Orthogonal DNFs). *A Boolean function φ is in orthogonal disjunctive normal form (orthogonal DNF) (Crama & Hammer, 2011) if it is a disjunction of terms $T_1 \vee T_2 \vee \dots \vee T_m$, such that for every pair of distinct terms T_i and T_j ($i \neq j$):*

$$T_i \wedge T_j \models \perp$$

Equivalently, no assignment satisfies two distinct terms simultaneously.

The proof follows the same structure as Theorem 1: it suffices to show that v_{suff}^g is polynomial-time computable for every candidate set S .

For a fixed set S , let $\mathcal{C}^S T_i$ denote the constraints imposed by term T_i on the coordinates in S . We enumerate all ordered term pairs (T_i, T_j) . If the pair induces inconsistent constraints on S (i.e., $\mathcal{C}_{T_i}^S \wedge \mathcal{C}_{T_j}^S \models \perp$), then its contribution to v_{suff}^g is zero and the pair can be skipped.

For each *consistent* term pair (T_i, T_j) , compute $\Pr(\mathbf{x}_S \in \mathcal{C}_{T_i, T_j}^S)$, where $\mathcal{C}_{T_i, T_j}^S = \mathcal{C}_{T_i}^S \cap \mathcal{C}_{T_j}^S$. The pair contributes the following product to v_{suff}^g :

$$\Pr(\mathbf{x}_S \in \mathcal{C}_{T_i, T_j}^S) \cdot \Pr(\mathbf{x}_{\bar{S}} \in \mathcal{C}_{T_i}^{\bar{S}}) \cdot \Pr(\mathbf{x}_{\bar{S}} \in \mathcal{C}_{T_j}^{\bar{S}}).$$

Under feature independence, each probability factorizes as $\Pr(\mathbf{x}_S) = \prod_{i \in S} p(x_i)$. There are at most m^2 term pairs, so evaluating all nonzero contributions computes v_{suff}^g in polynomial time under an independent distribution $\mathcal{D}$. □

O DISCLOSURE: USAGE OF LLMs

An LLM was used exclusively for language-level assistance, including grammar correction, typo correction, and clarity improvements. It was not used to generate research ideas, design the study, analyze data, or interpret results; those tasks were carried out solely by the authors.